

# MOVING CHARGES AND MAGNETISM

Chapter Four  
Class - XII

Manish Joshi

D.A.V. Lohaghat

June 30, 2025



## Learning Objectives:

- Concept of magnetic field, Oersted's experiment.



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# Oersted's Experiment

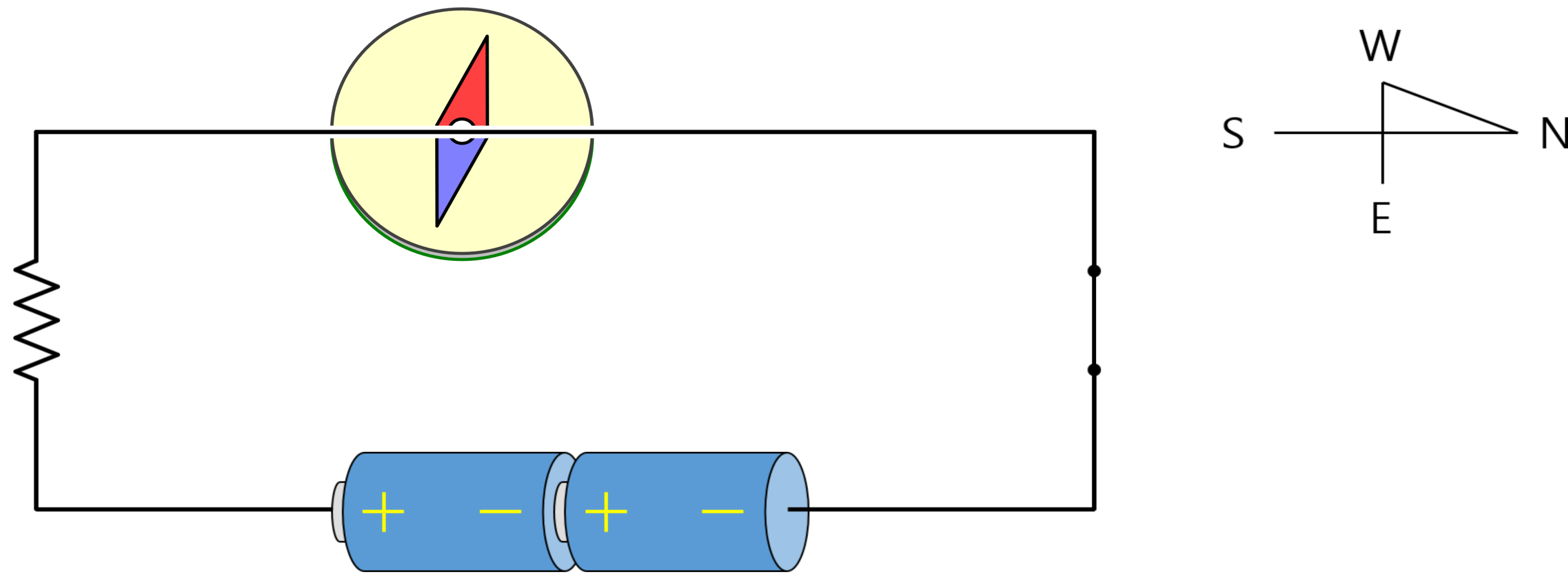


Figure: Oersted's Experiment Setup.  
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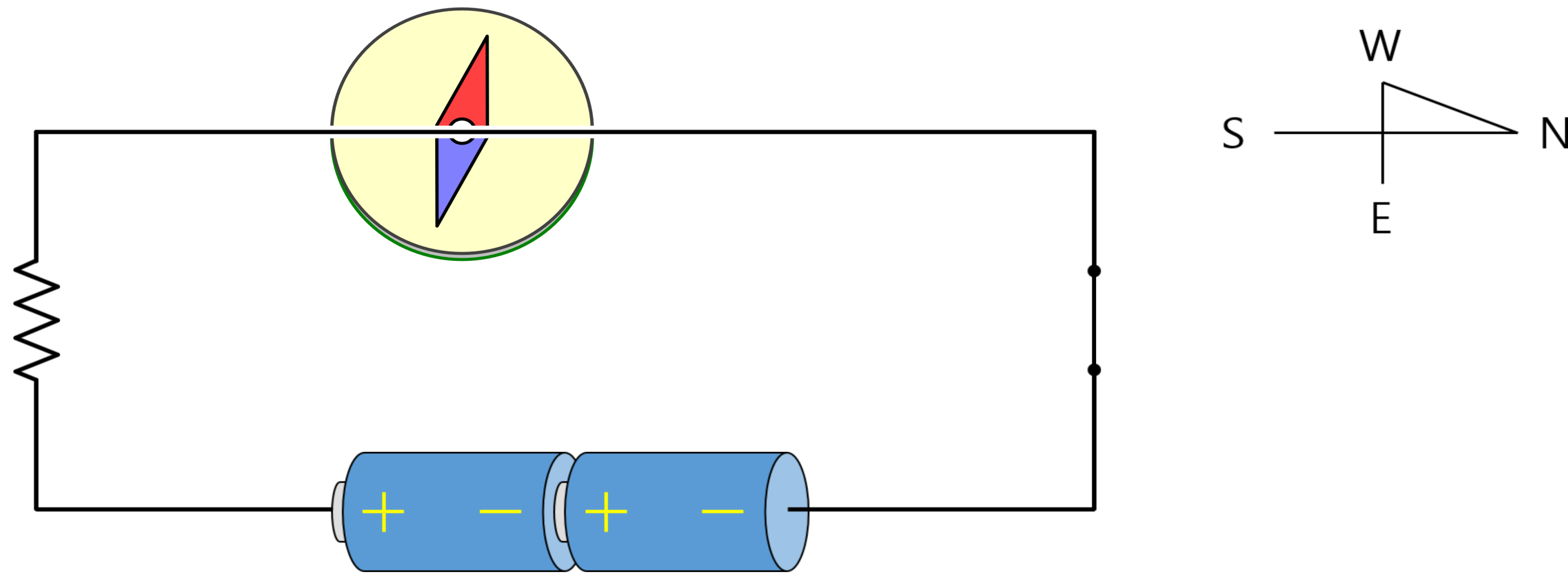


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- **Oersted experimented** with this phenomenon several times, **revealing a close relationship between electrical and magnetic phenomena**.



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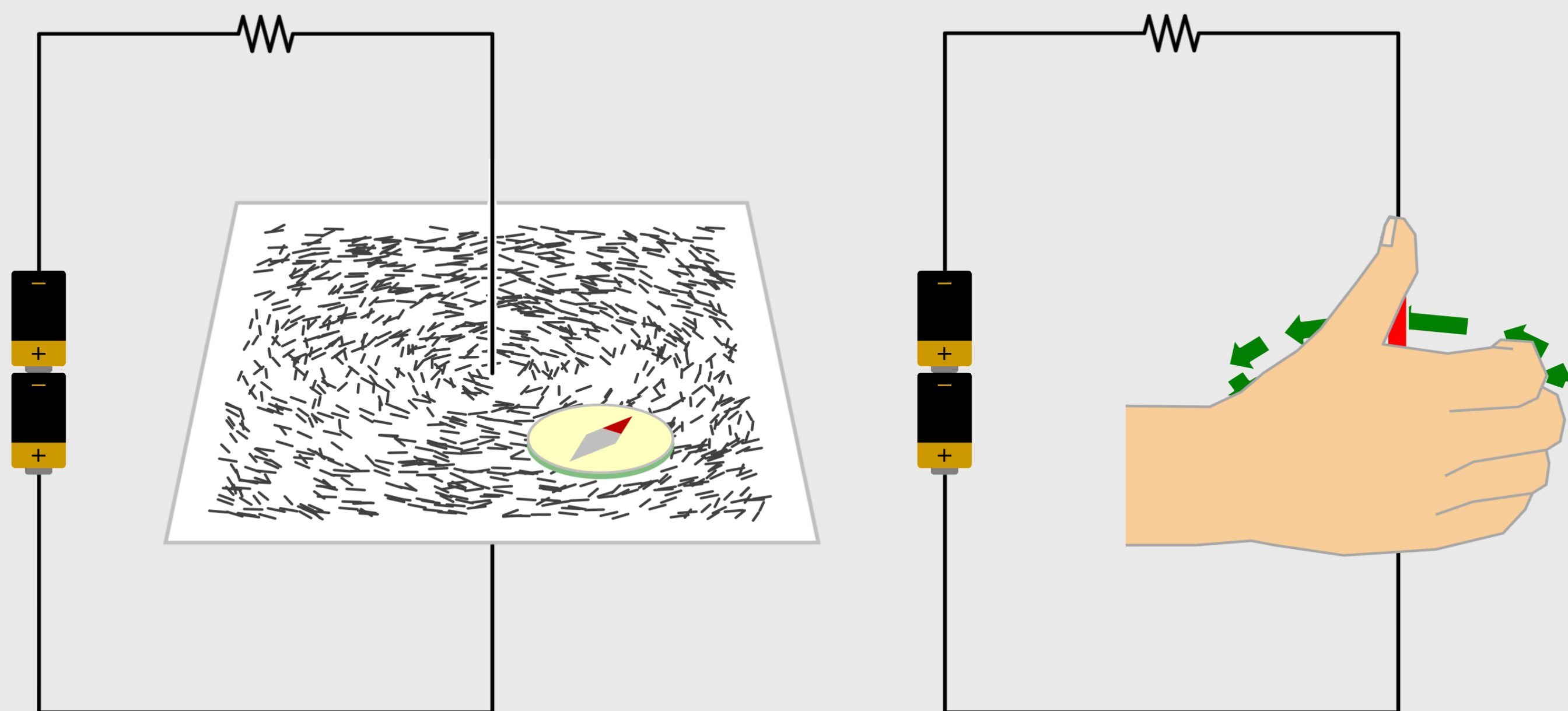


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## Observations and Results:

- He found that the **alignment of the needle is tangential to an imaginary circle** which has the straight wire as its center and has its plane perpendicular to the wire.



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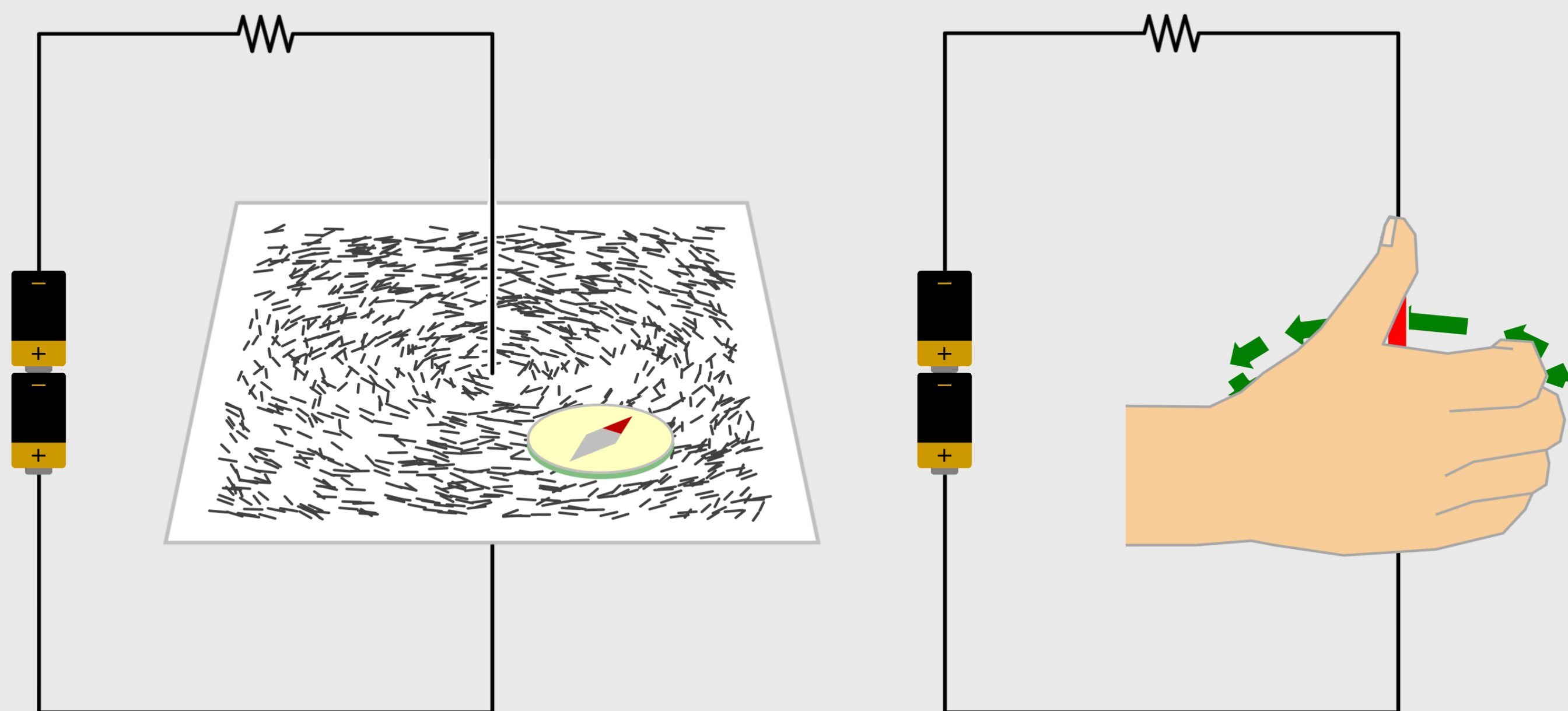


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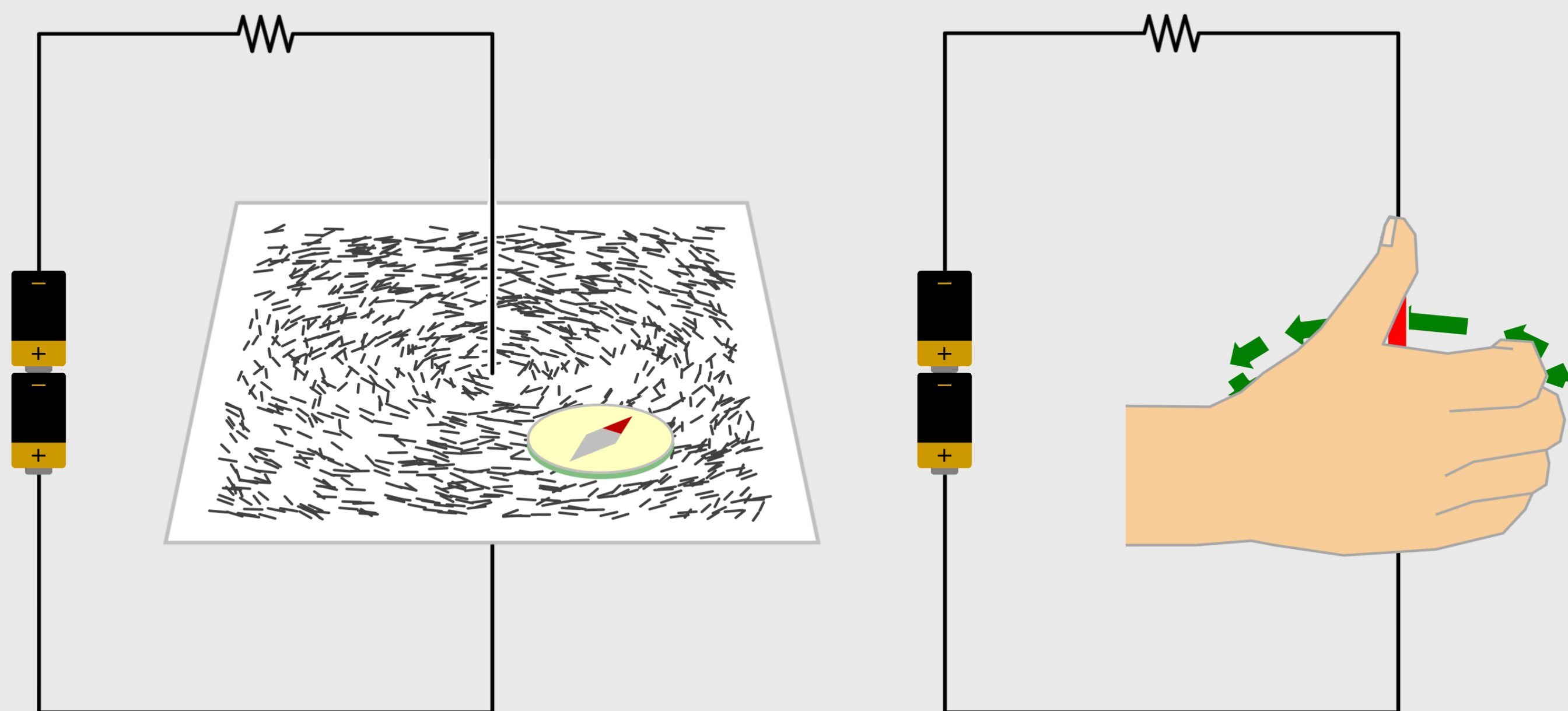


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- **Reversing** the direction of the **current** **reverses the orientation of the needle.**
- The **deflection increases on increasing the current** or **bringing the needle closer to the wire.**



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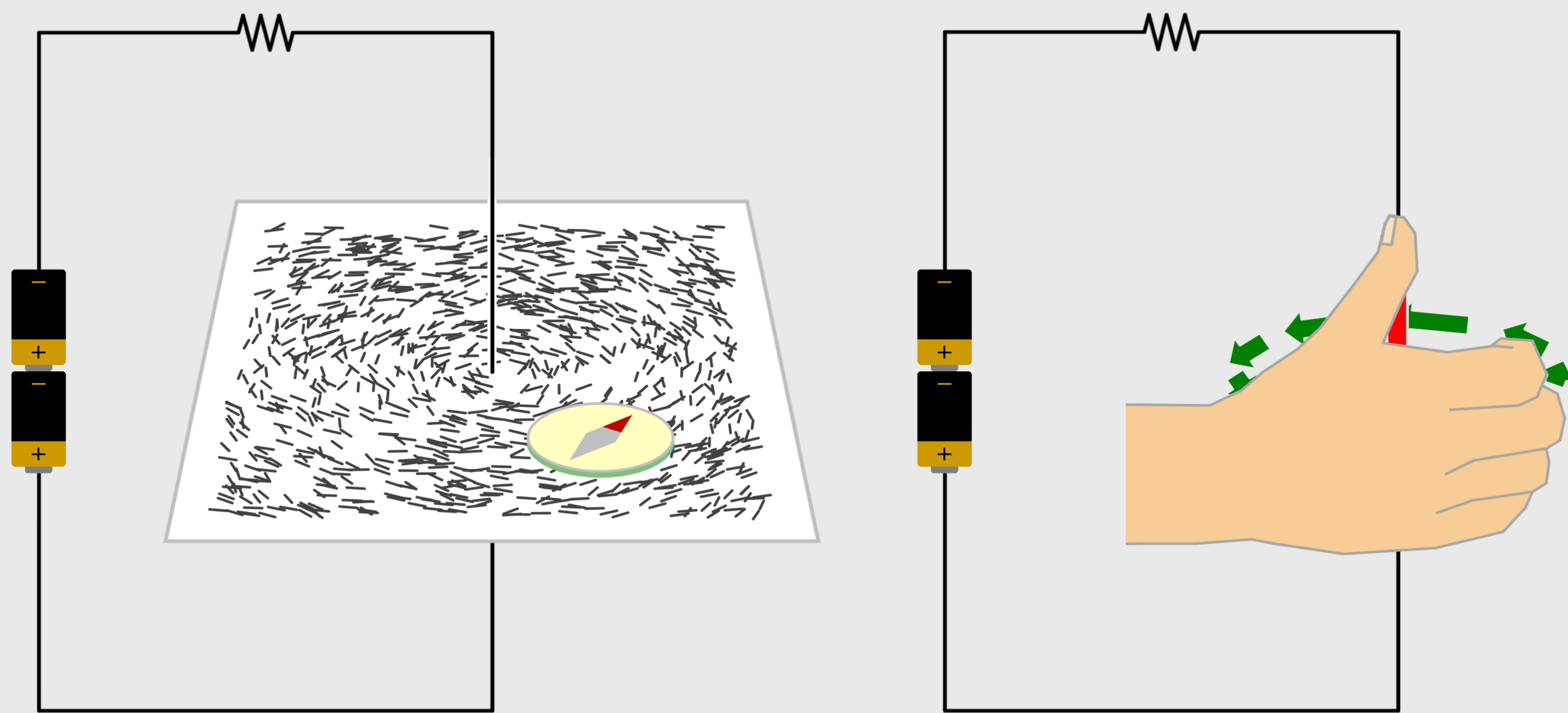


Figure: Magnetic Field around a Wire and its direction is give by right hand thumb rule. Source : Java Lab

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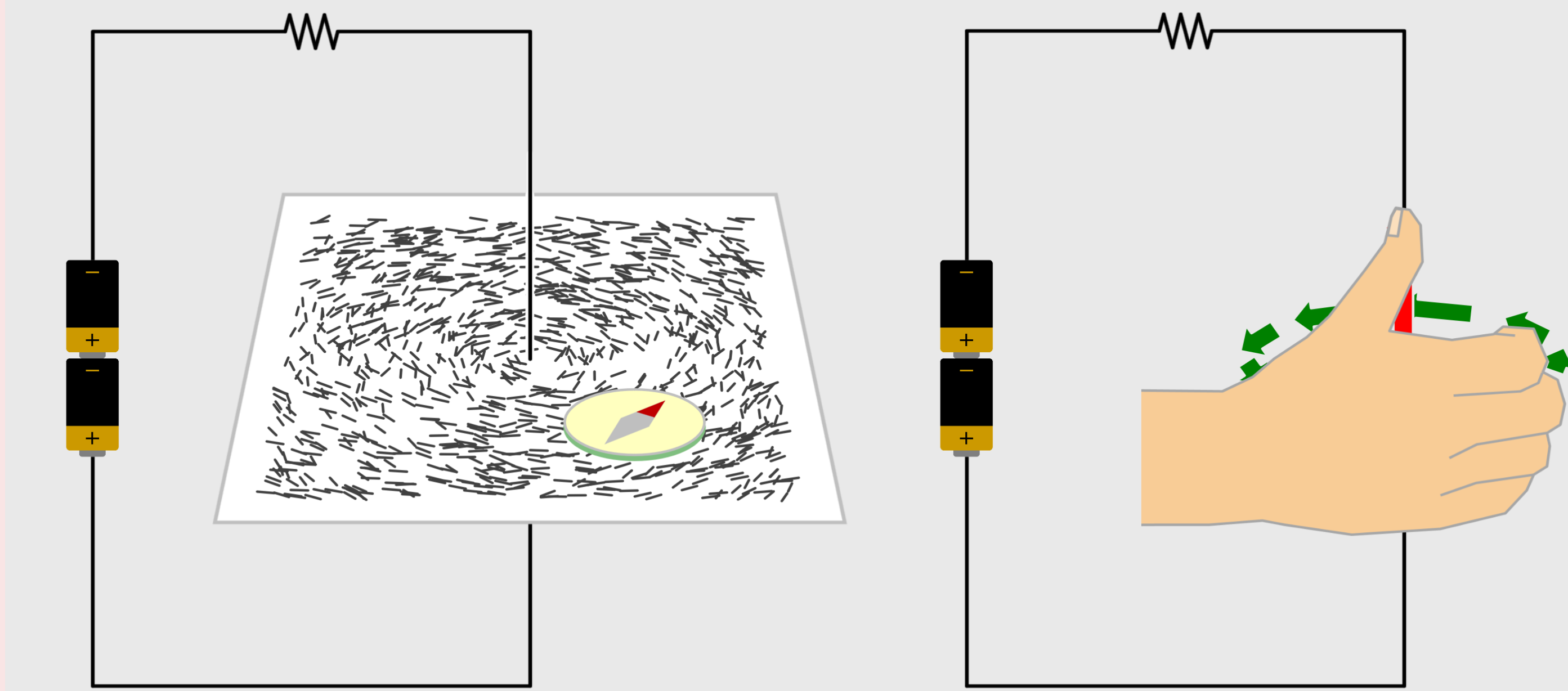


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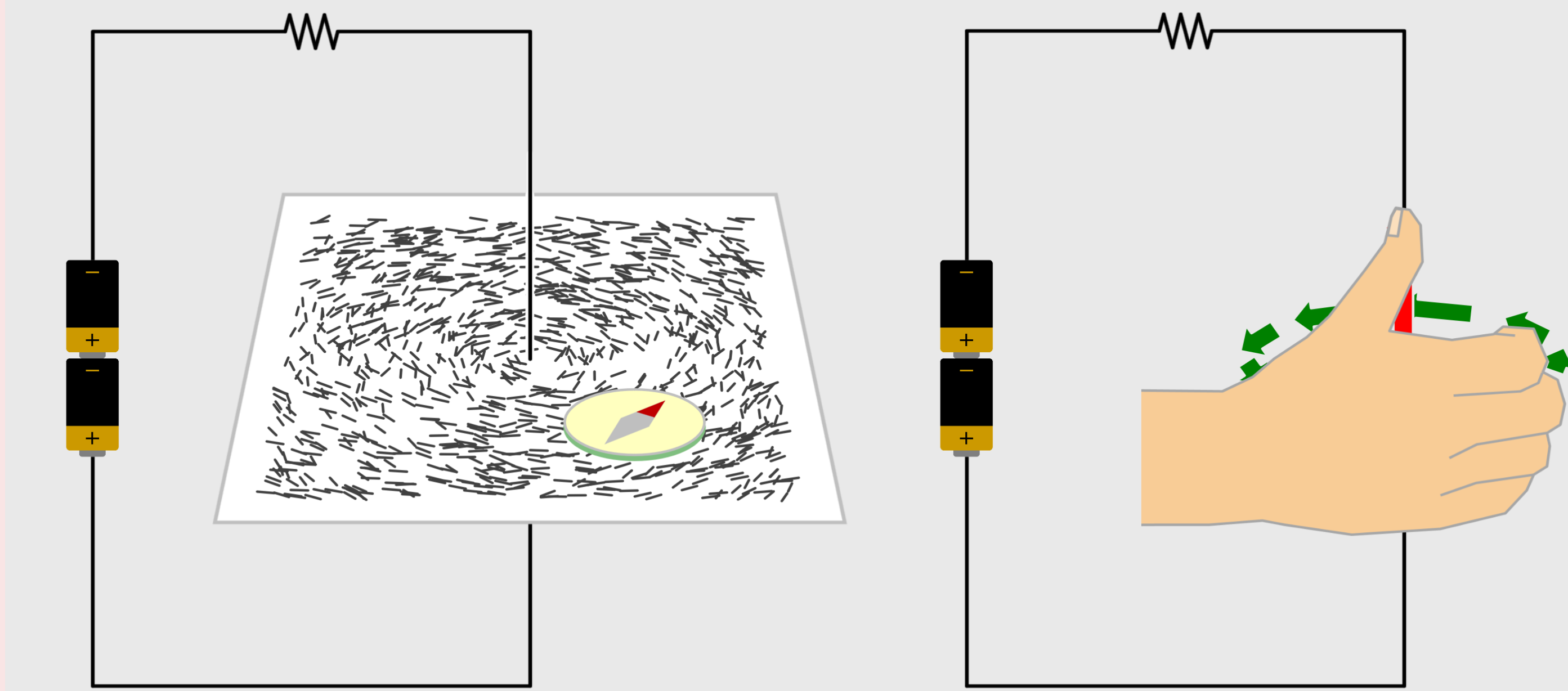


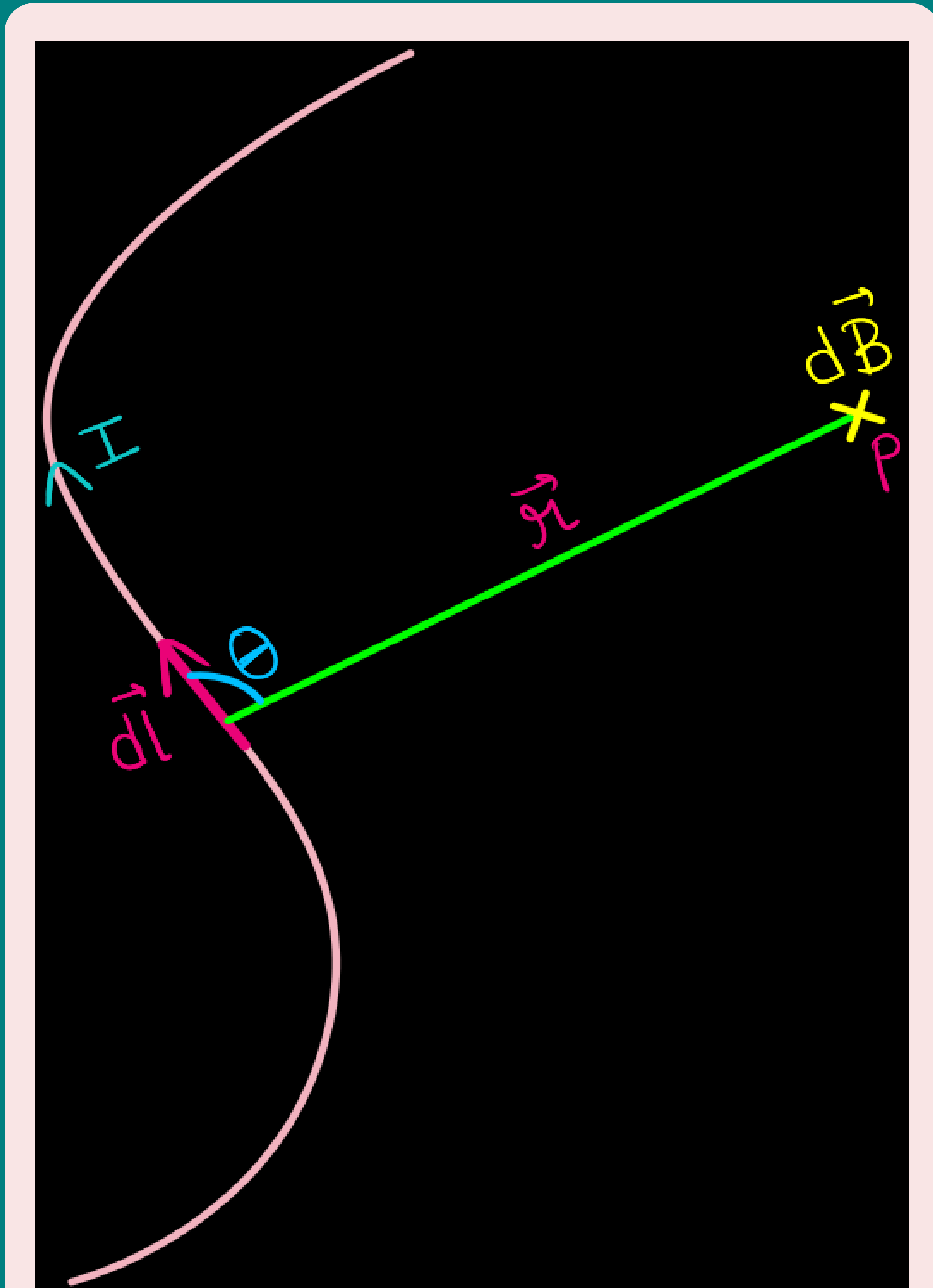
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- **Oersted concluded** that **moving charges or currents produced a magnetic field** in the surrounding space.



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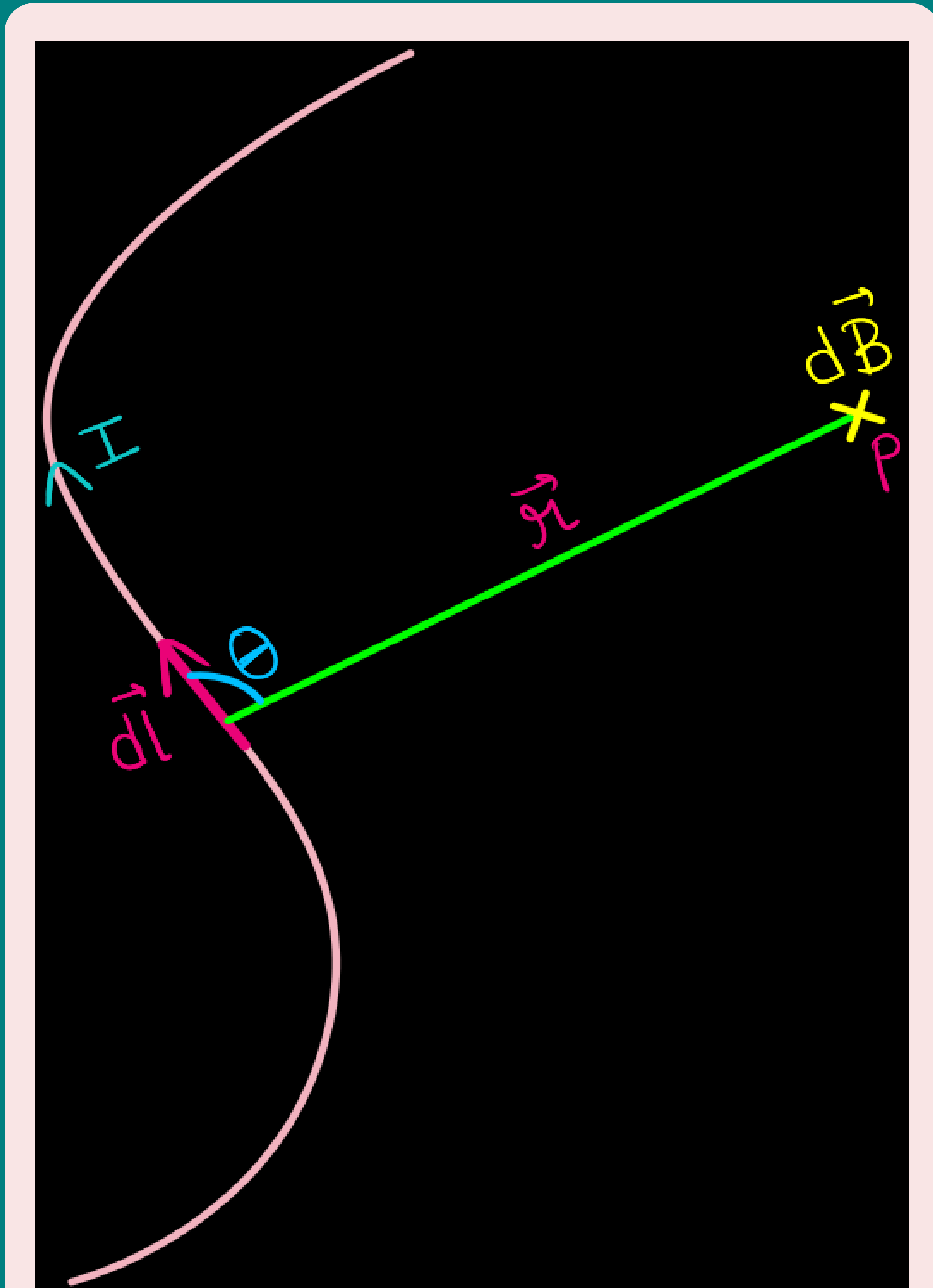


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- Figure shows a finite conductor carrying **current  $I$** .



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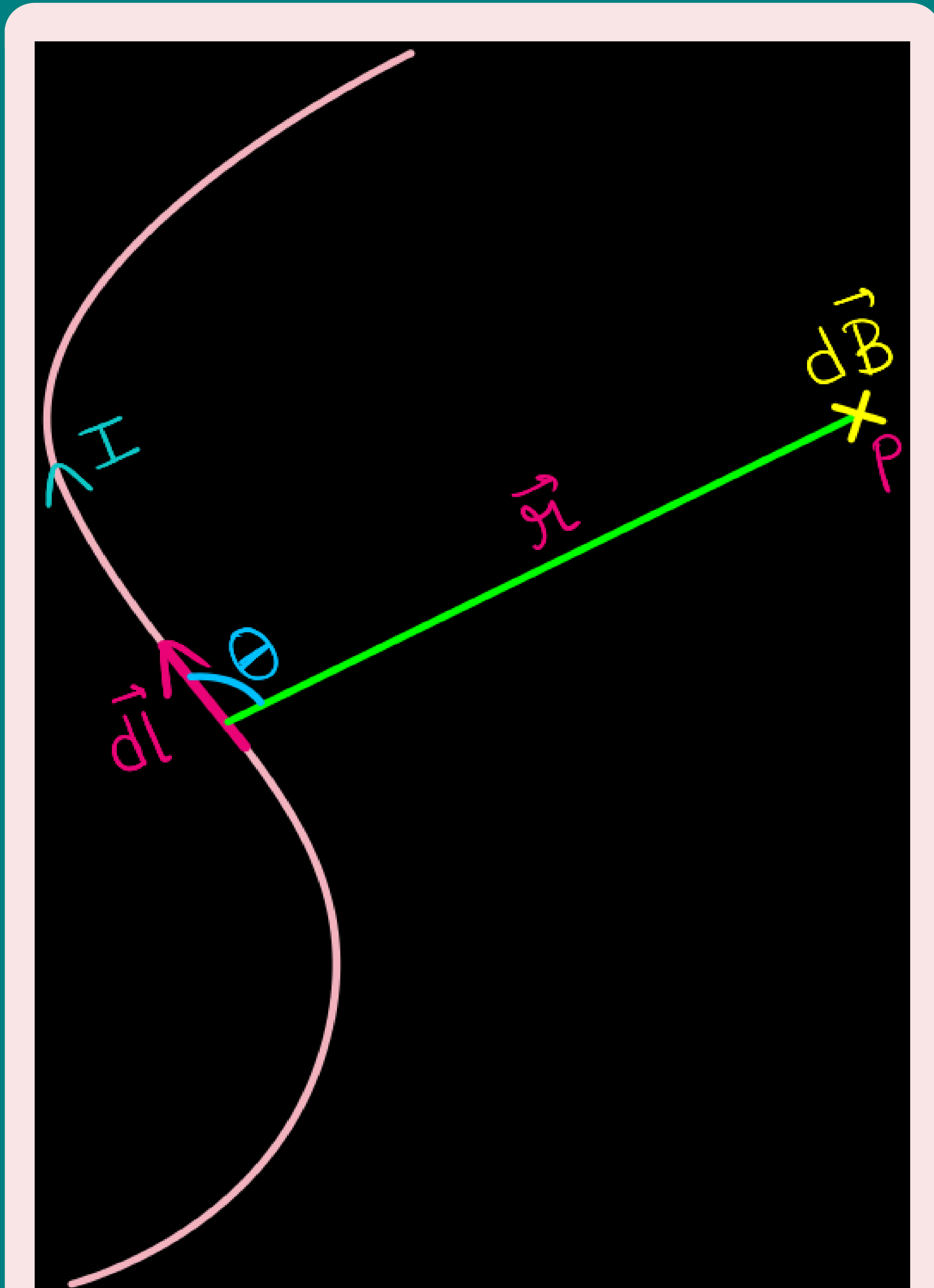


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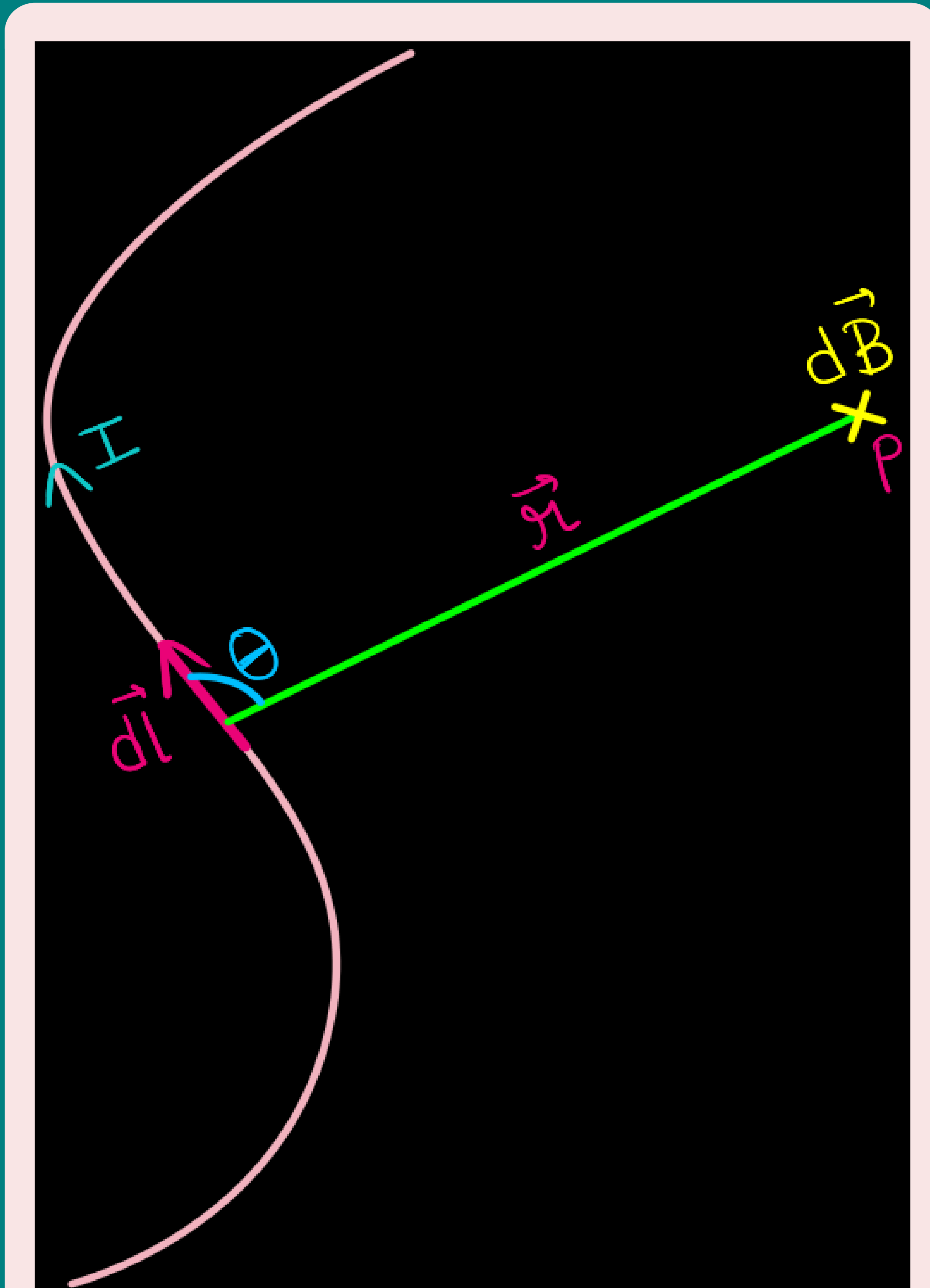


## Magnetic Field Due to a Current Element

- Figure shows a finite conductor carrying **current  $I$** .
- Consider an **infinitesimal element  $d\vec{l}$**  of the conductor.
- The **magnetic field  $d\vec{B}$**  due to this element is to be determined **at a point  $P$**  which is at a **distance  $r$**  from it. Let  $\theta$  be the **angle between  $d\vec{l}$  and the displacement vector  $r$** .



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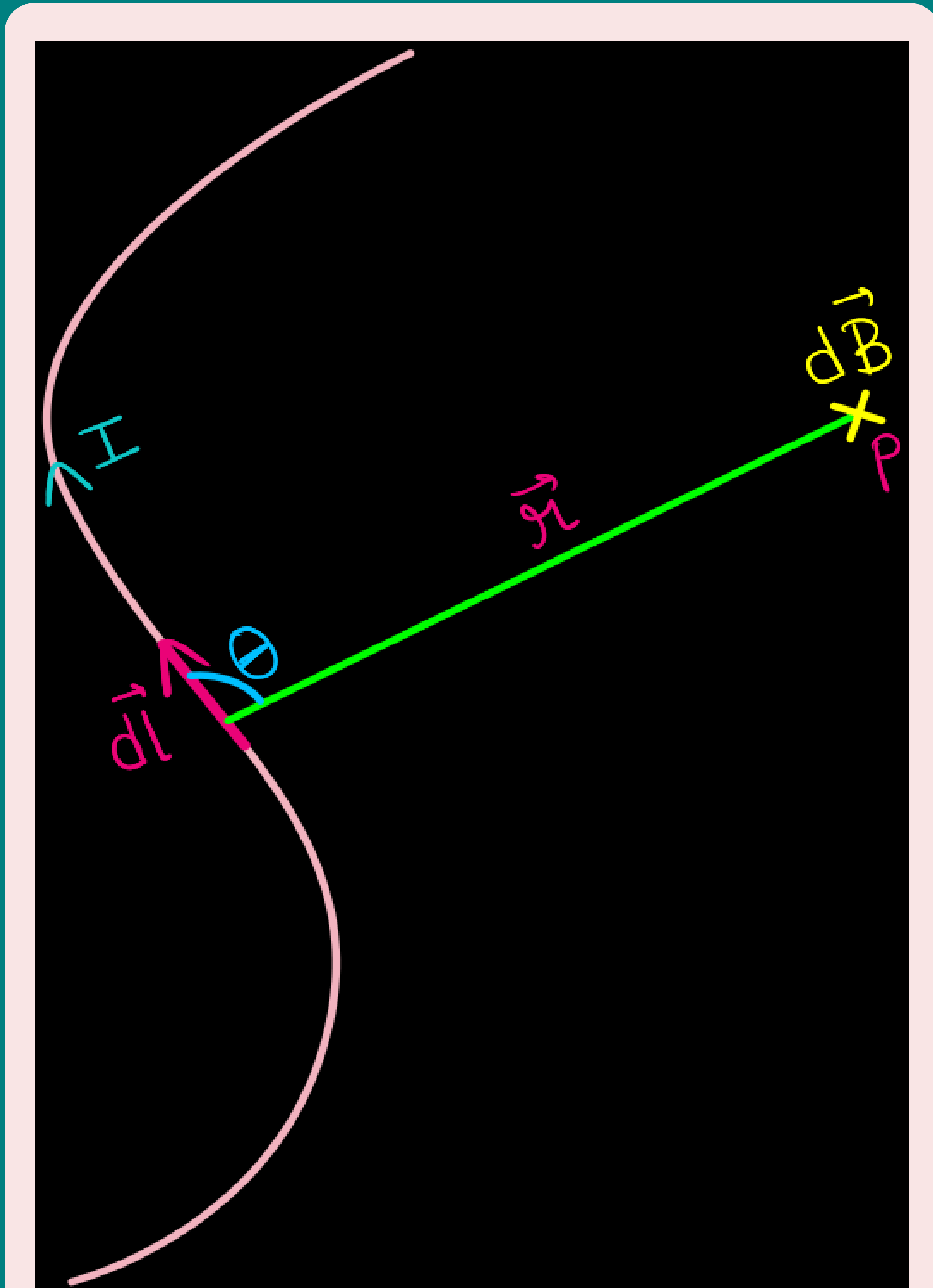


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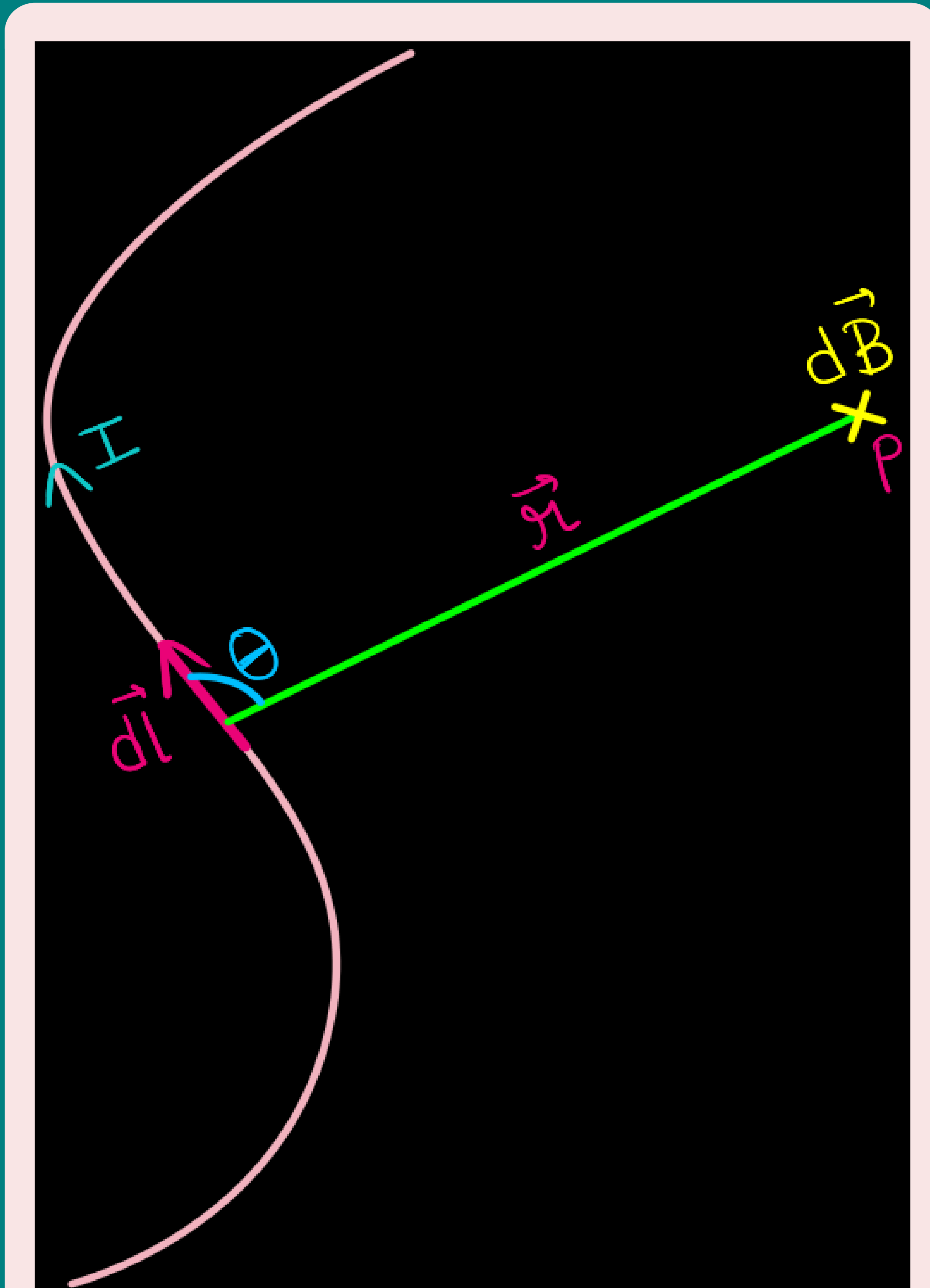
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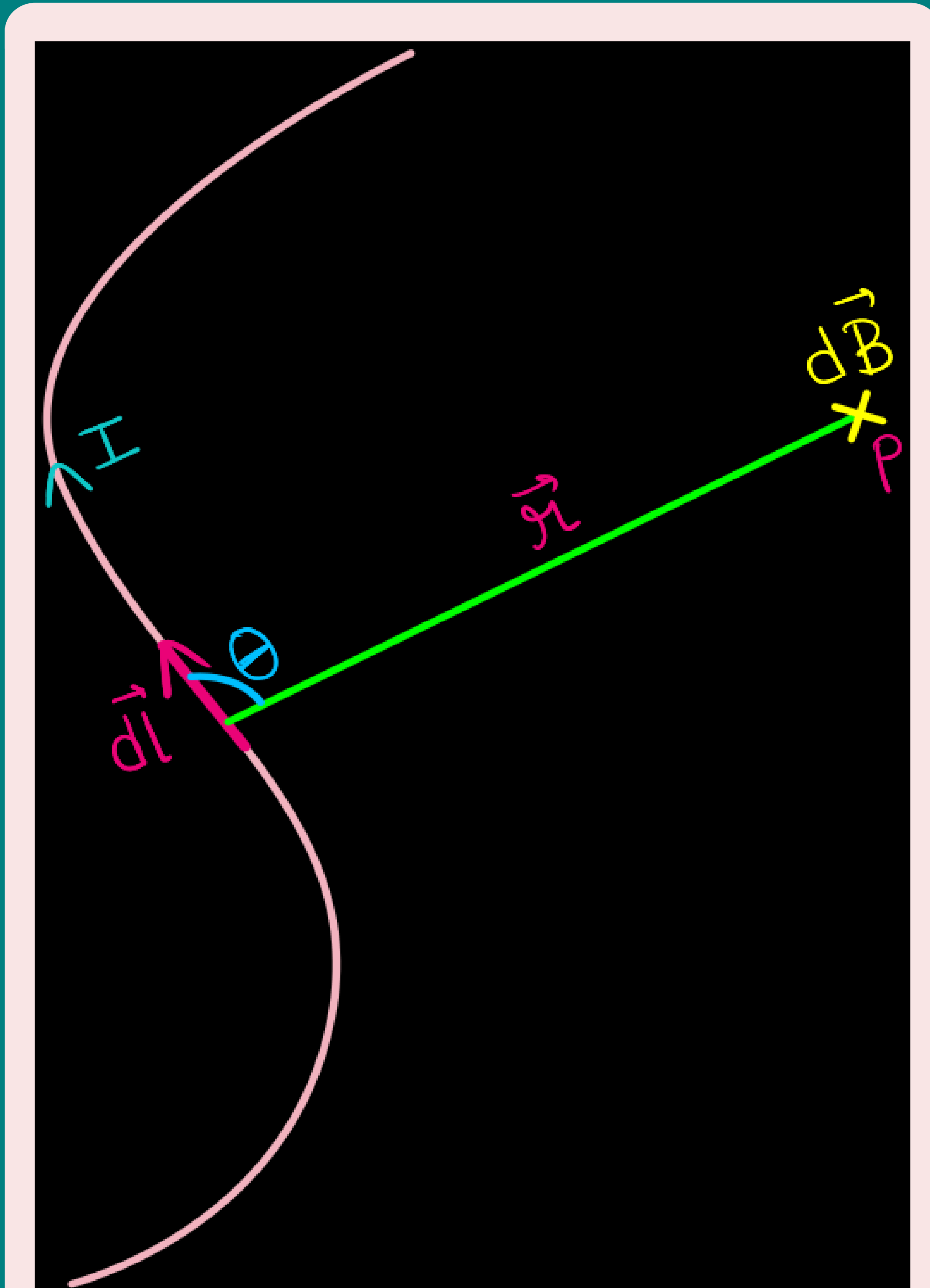
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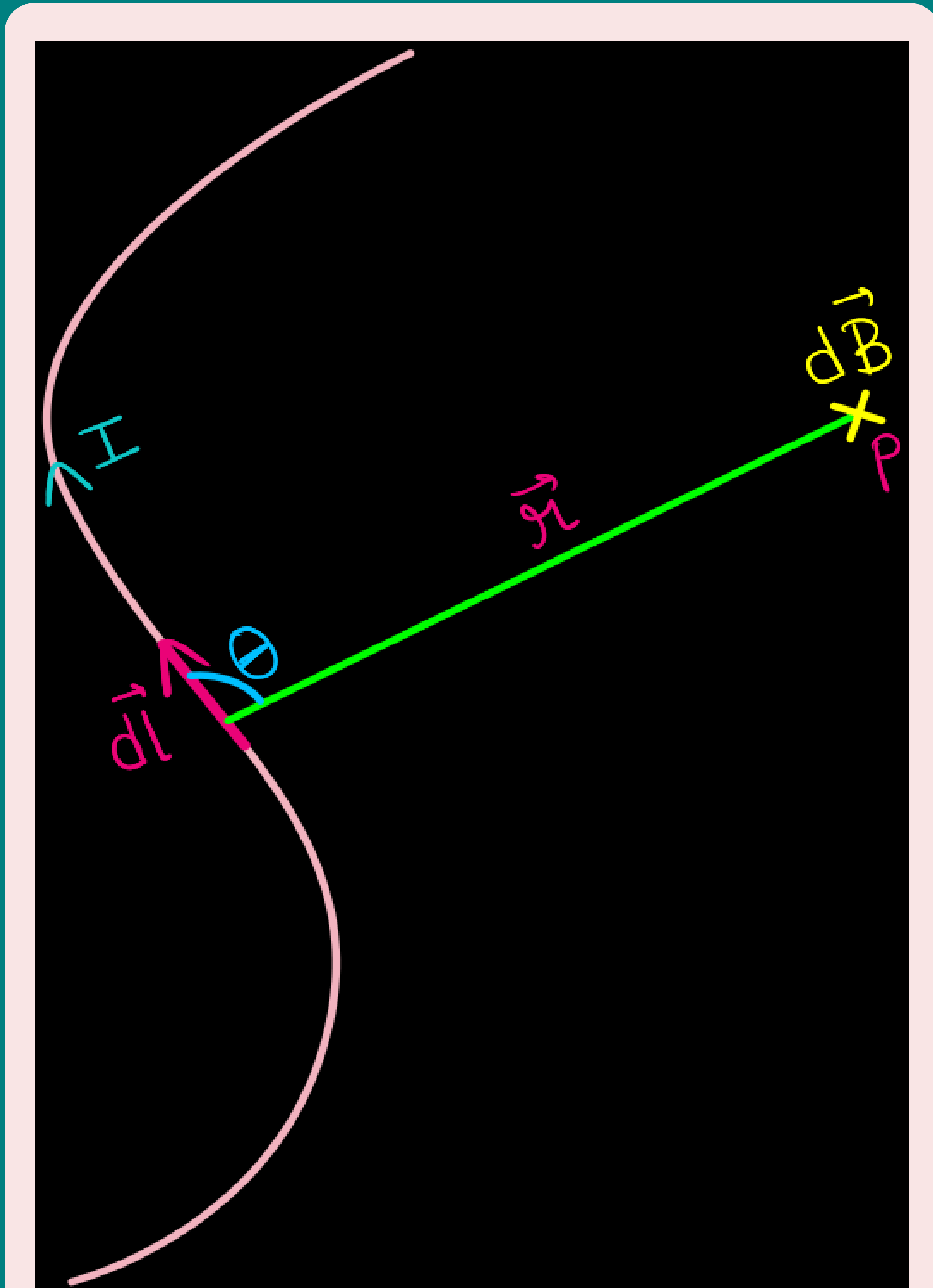
$$(d\vec{B} \propto d\vec{l}) \quad \text{and}$$

- 3 Inversely proportional to the square of the distance  $r$

$$\left( d\vec{B} \propto \frac{1}{r^2} \right).$$



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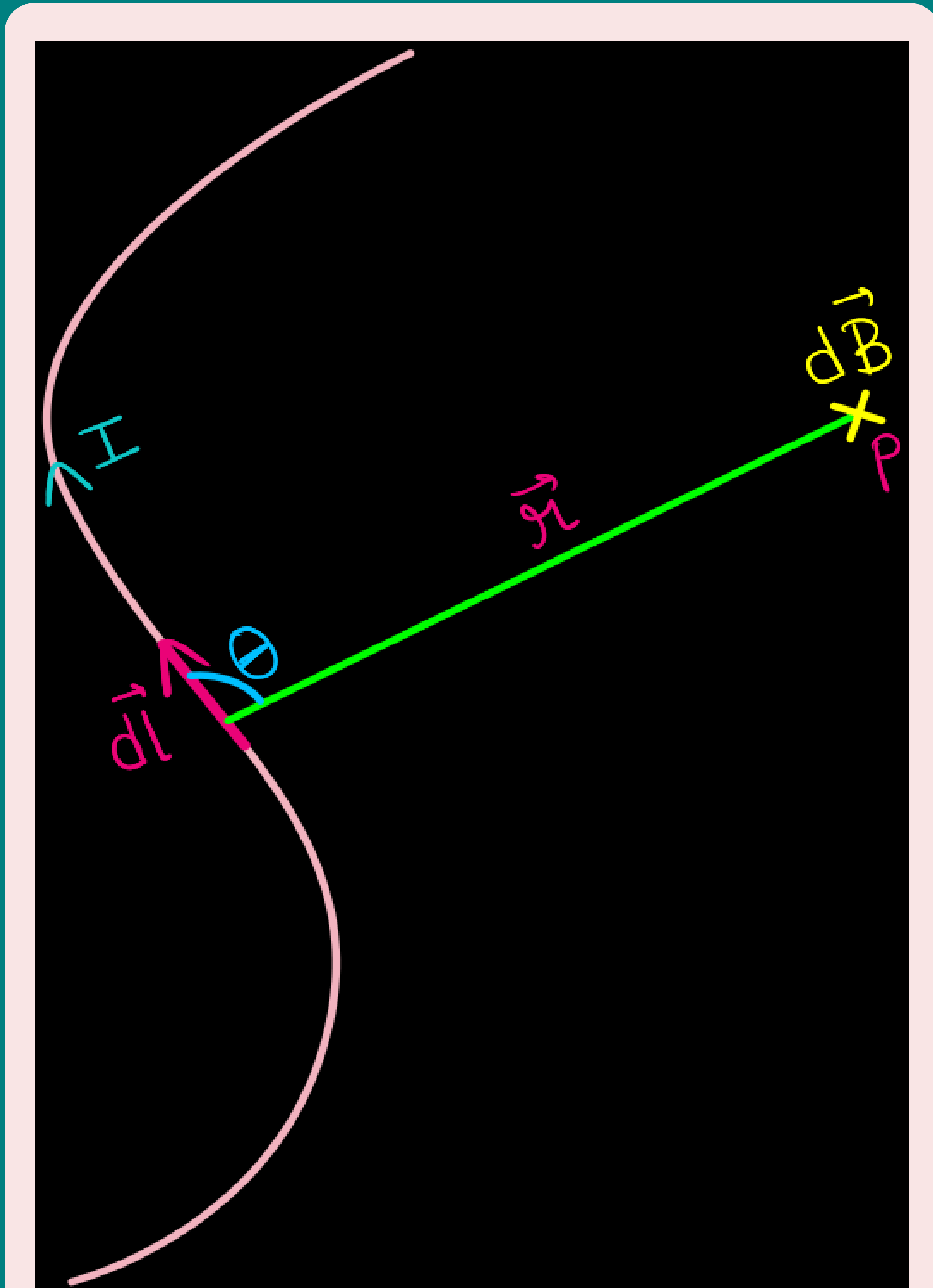


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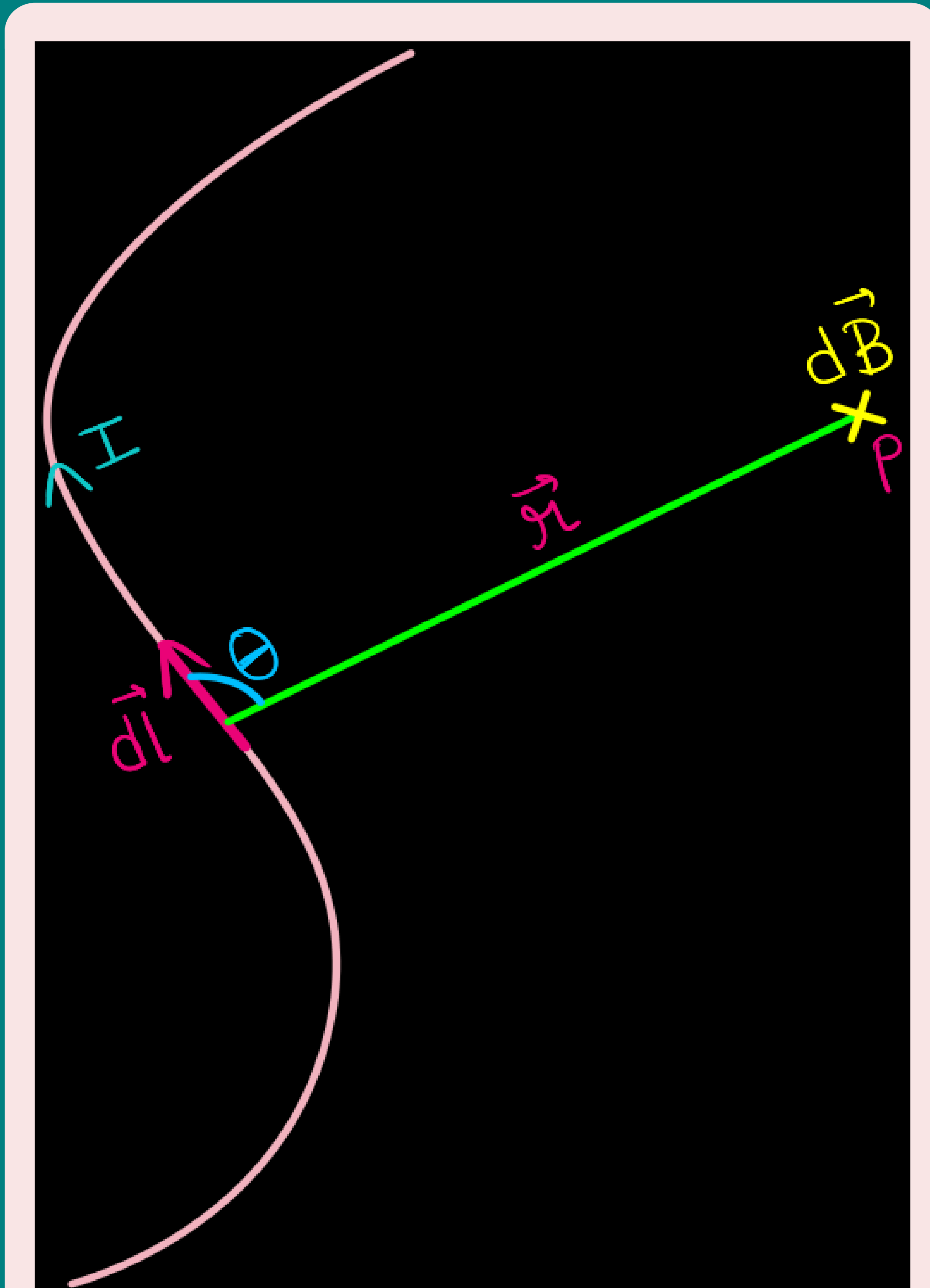
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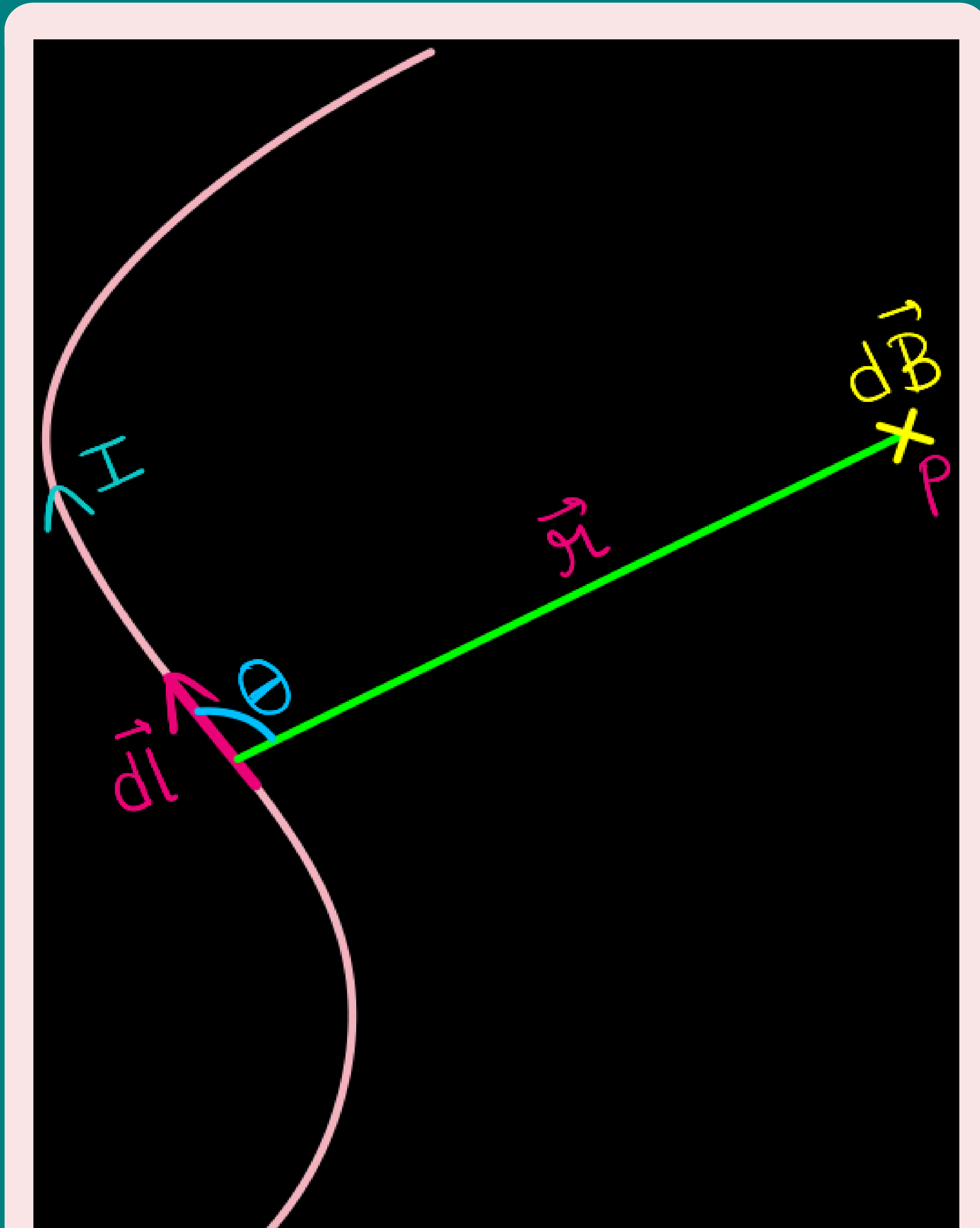
$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2}$$

or

$$|d\vec{B}| = \frac{\mu_0}{4\pi} \frac{I dl \sin \theta}{r^2}$$



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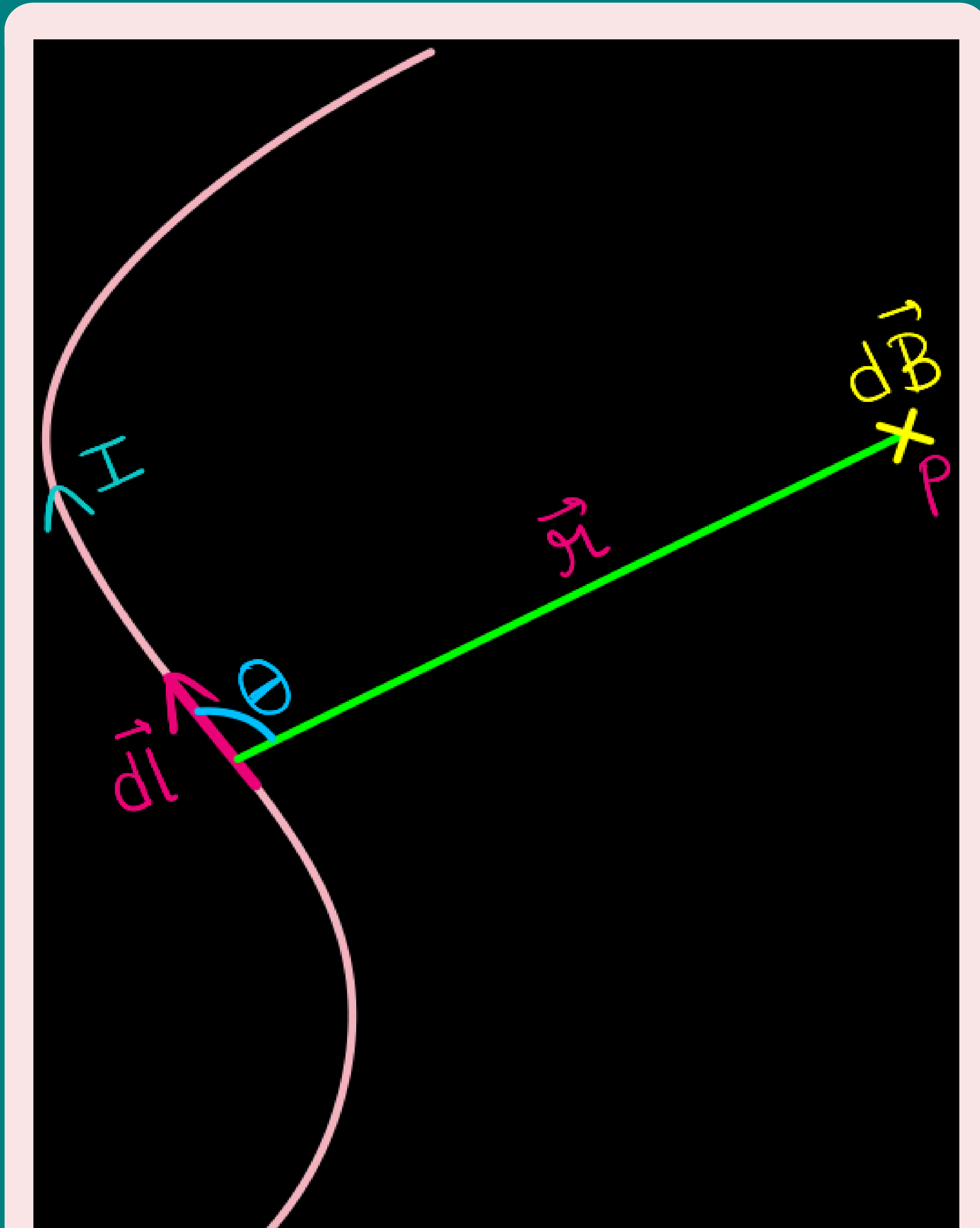


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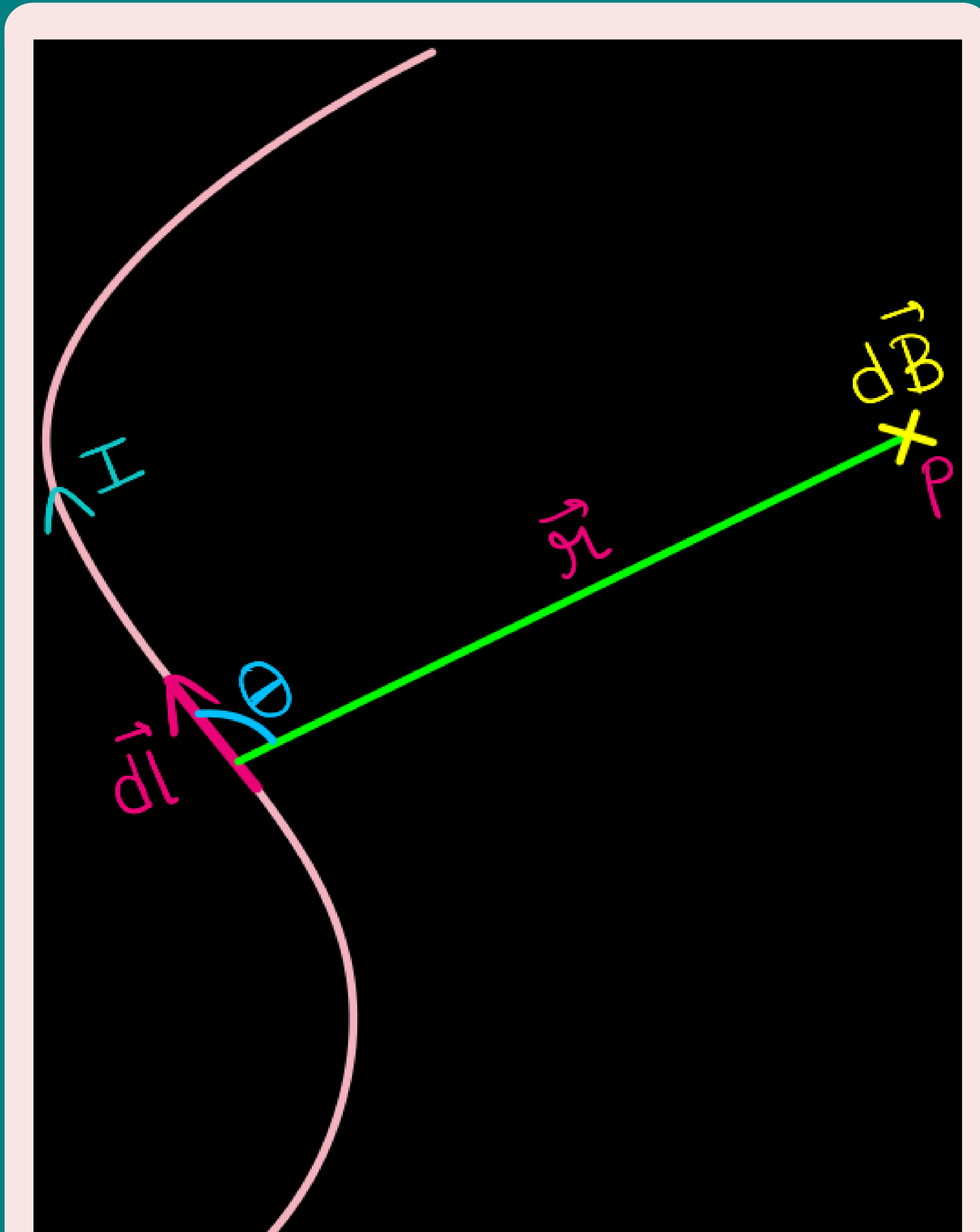
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- We call  $\mu_0$  the **permeability of free space (or vacuum)**.



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- There is an **angle dependence in the Biot-Savart law** which is **not present in the electrostatic case**.



# Relation between Permittivity of free space ( $\epsilon_0$ ) and the permeability of free space ( $\mu_0$ )

- There is an **interesting relation** between  $\epsilon_0$ , the **permittivity of free space**;  $\mu_0$ , the **permeability of free space**; and  $c$ , the **speed of light in vacuum**:



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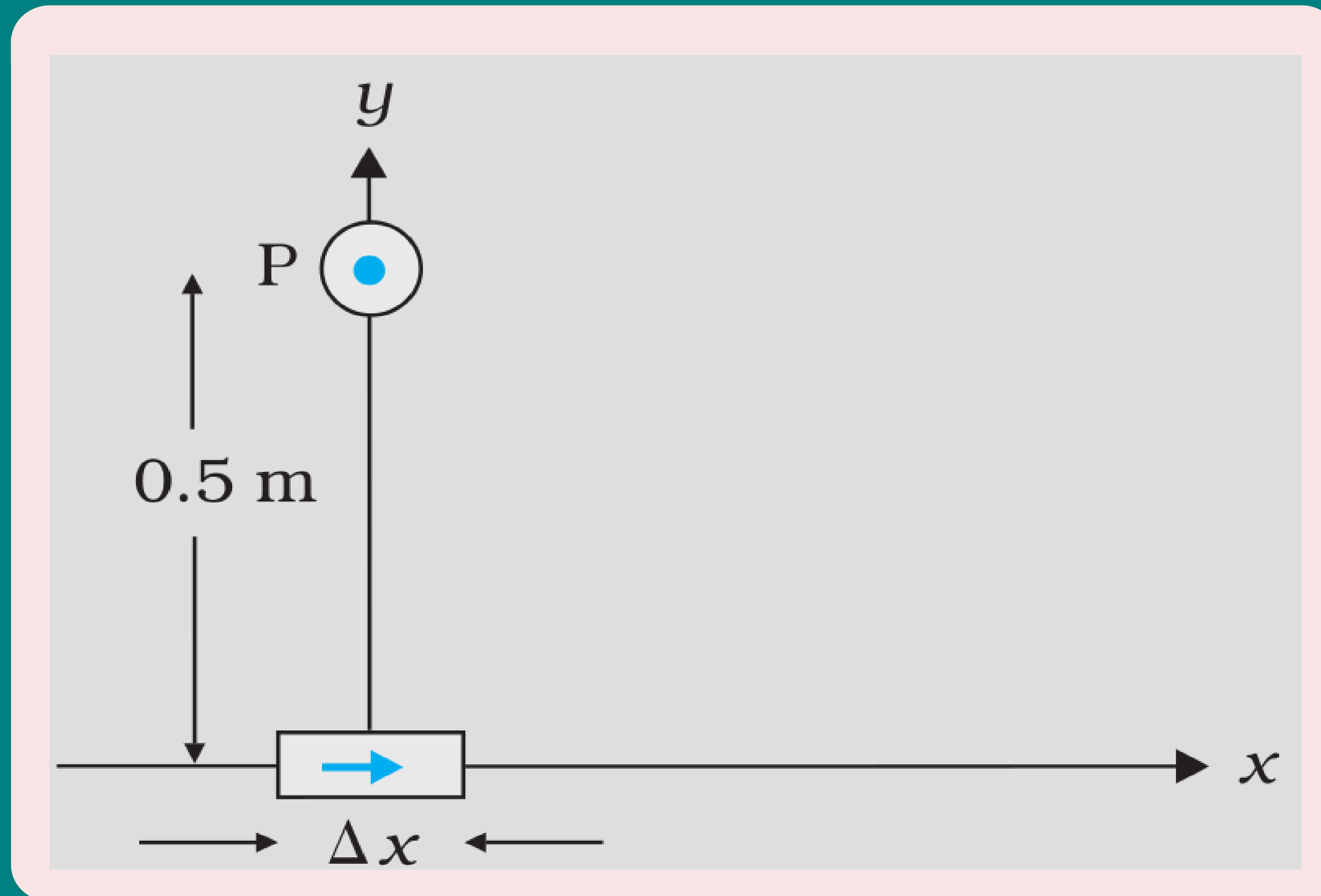
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## Example 4.5



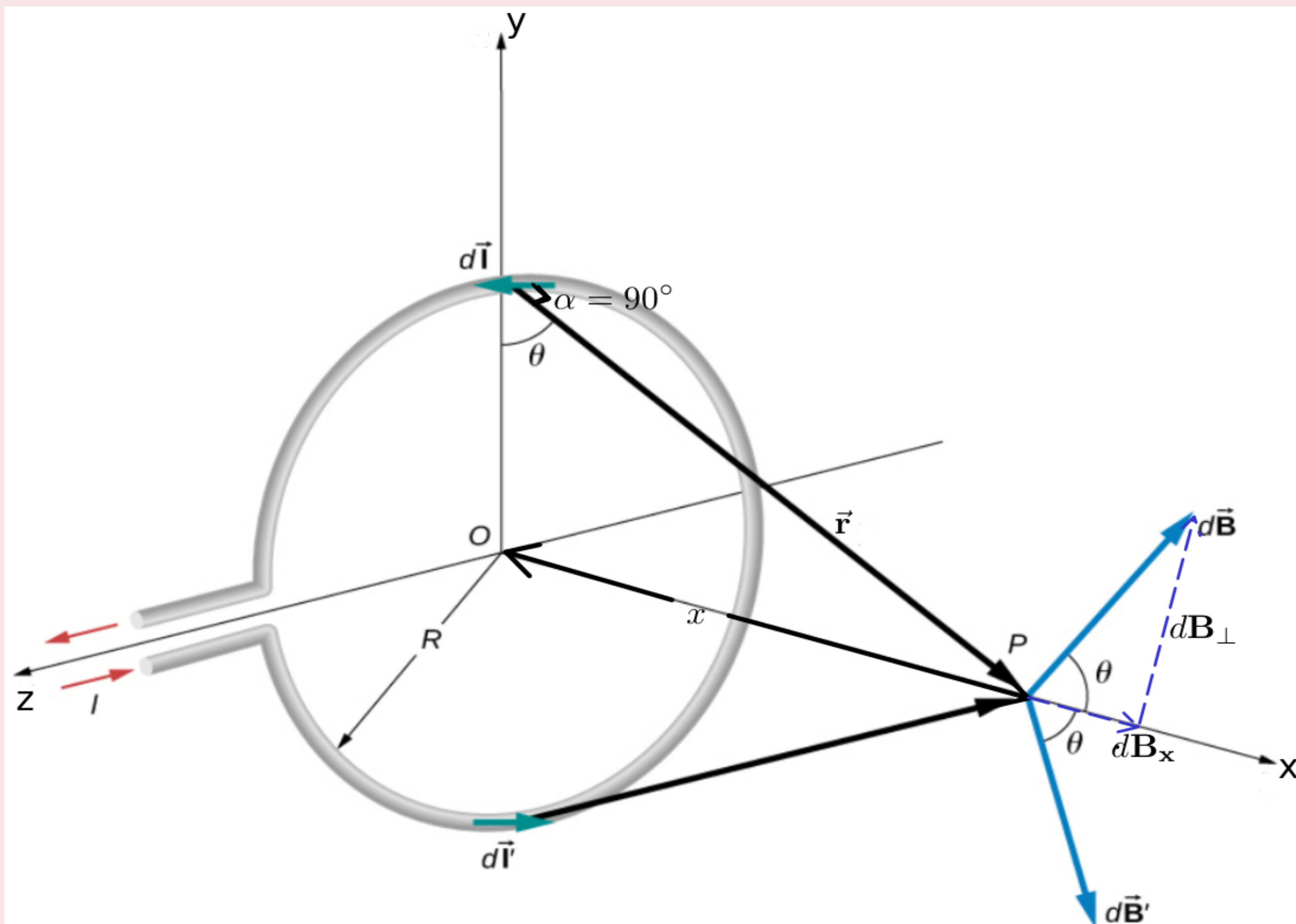
An element  $\Delta l = \Delta x \hat{i}$  is placed at the origin and carries a large current  $I = 10 \text{ A}$ . What is the magnetic field on the y-axis at a distance of  $0.5 \text{ m}$ .  $\Delta x = 1 \text{ cm}$ .



# Applications of Biot-Savart Law

## Magnetic Field On The Axis Of a Circular Current Loop

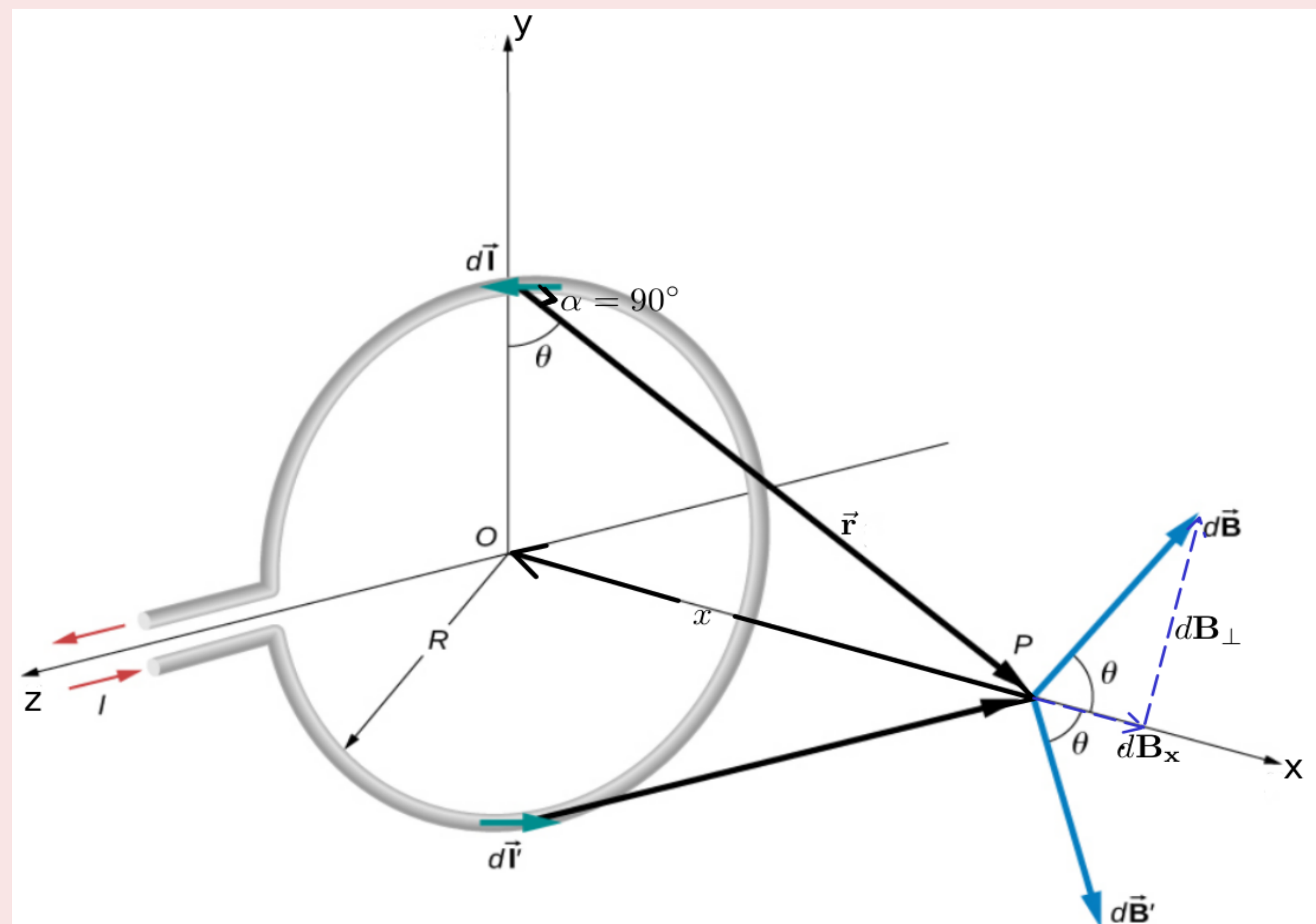
- Figure depicts a **circular loop carrying a steady current  $I$** .



**Figure 12.11** Determining the magnetic field at point  $P$  along the axis of a current-carrying loop of wire.



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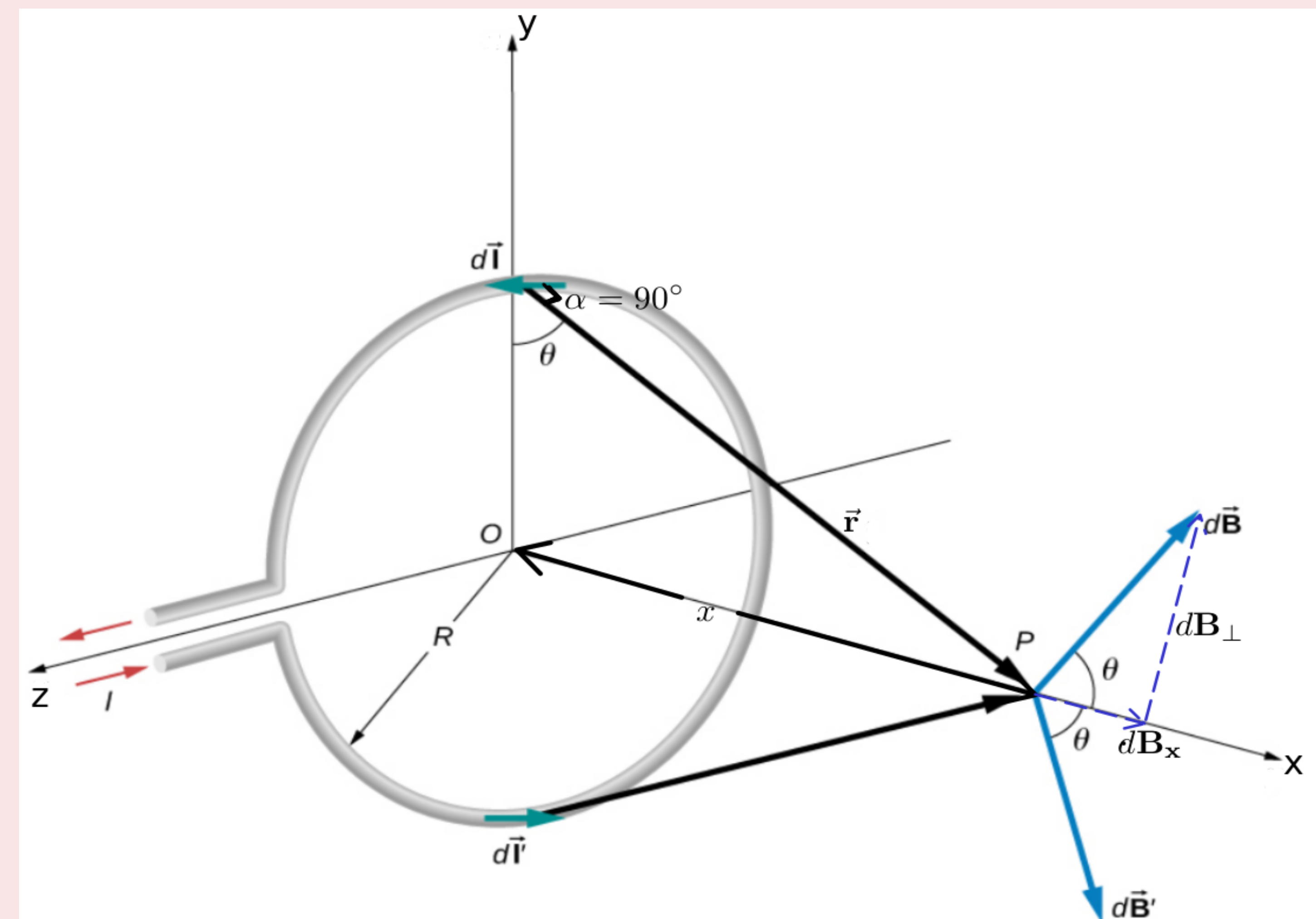


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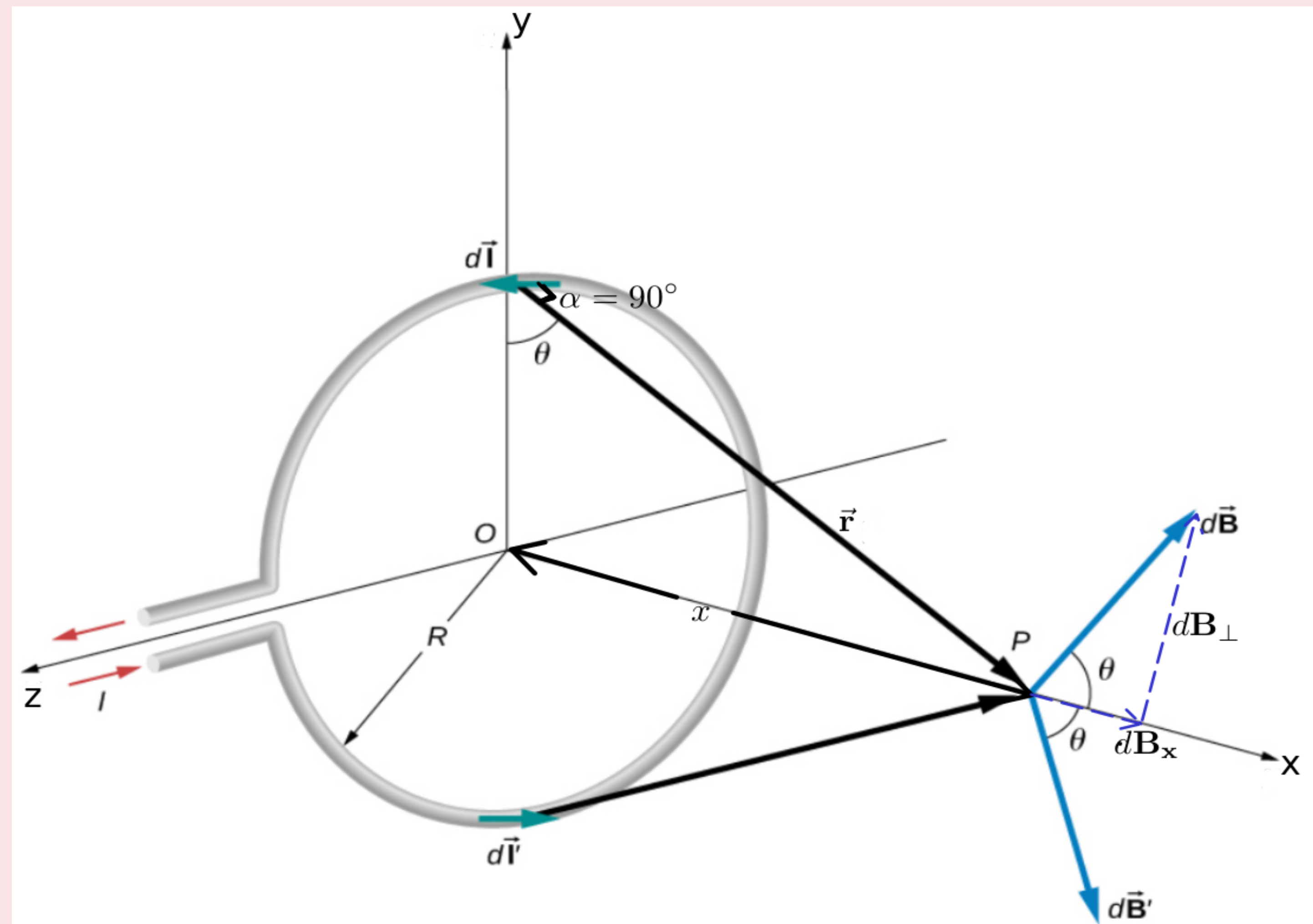


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- **Let  $x$  be the distance of  $P$  from the center  $O$  of the loop.**



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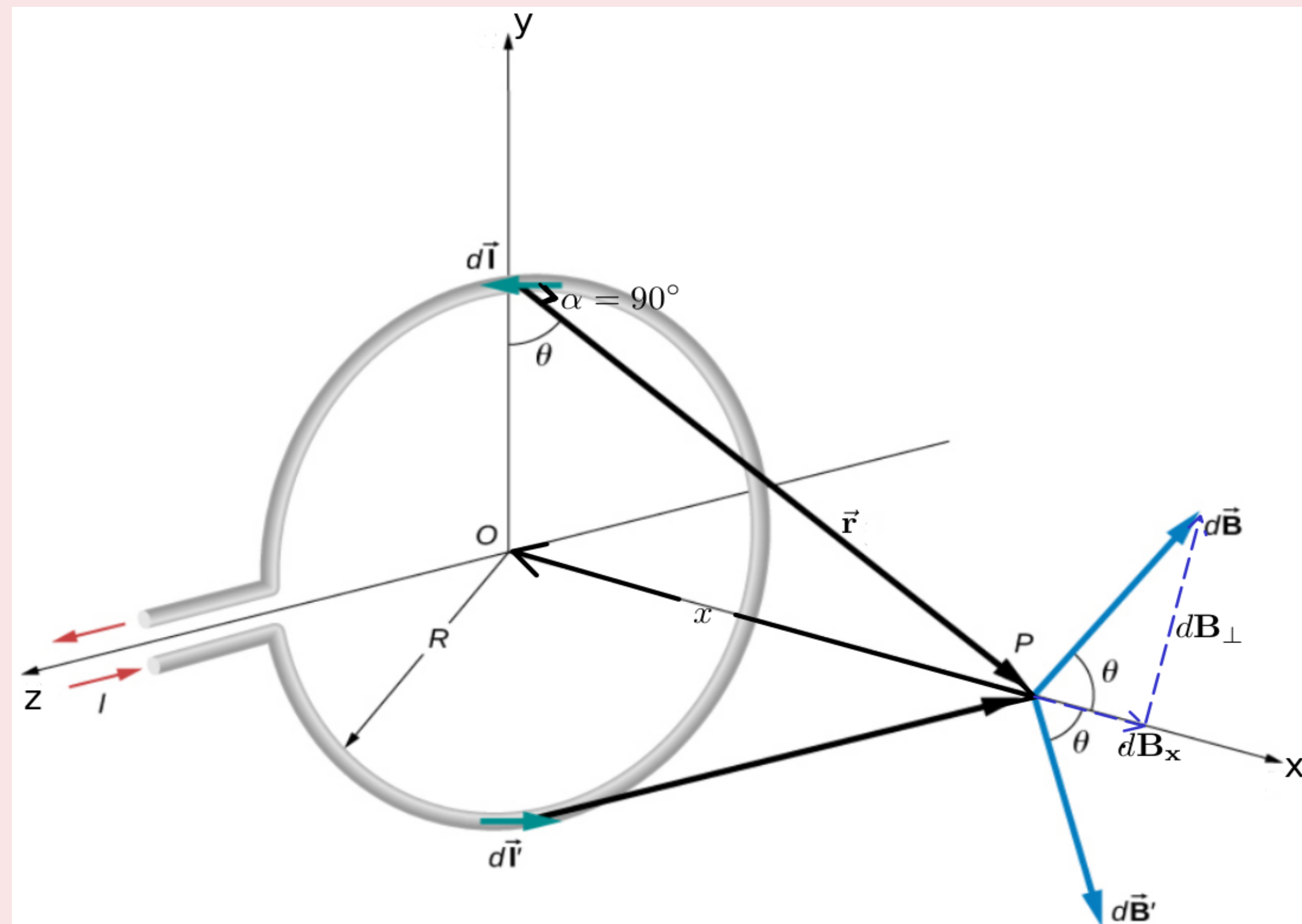


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$$dB = \frac{\mu_0}{4\pi} \frac{I dl \sin \alpha}{r^2}$$



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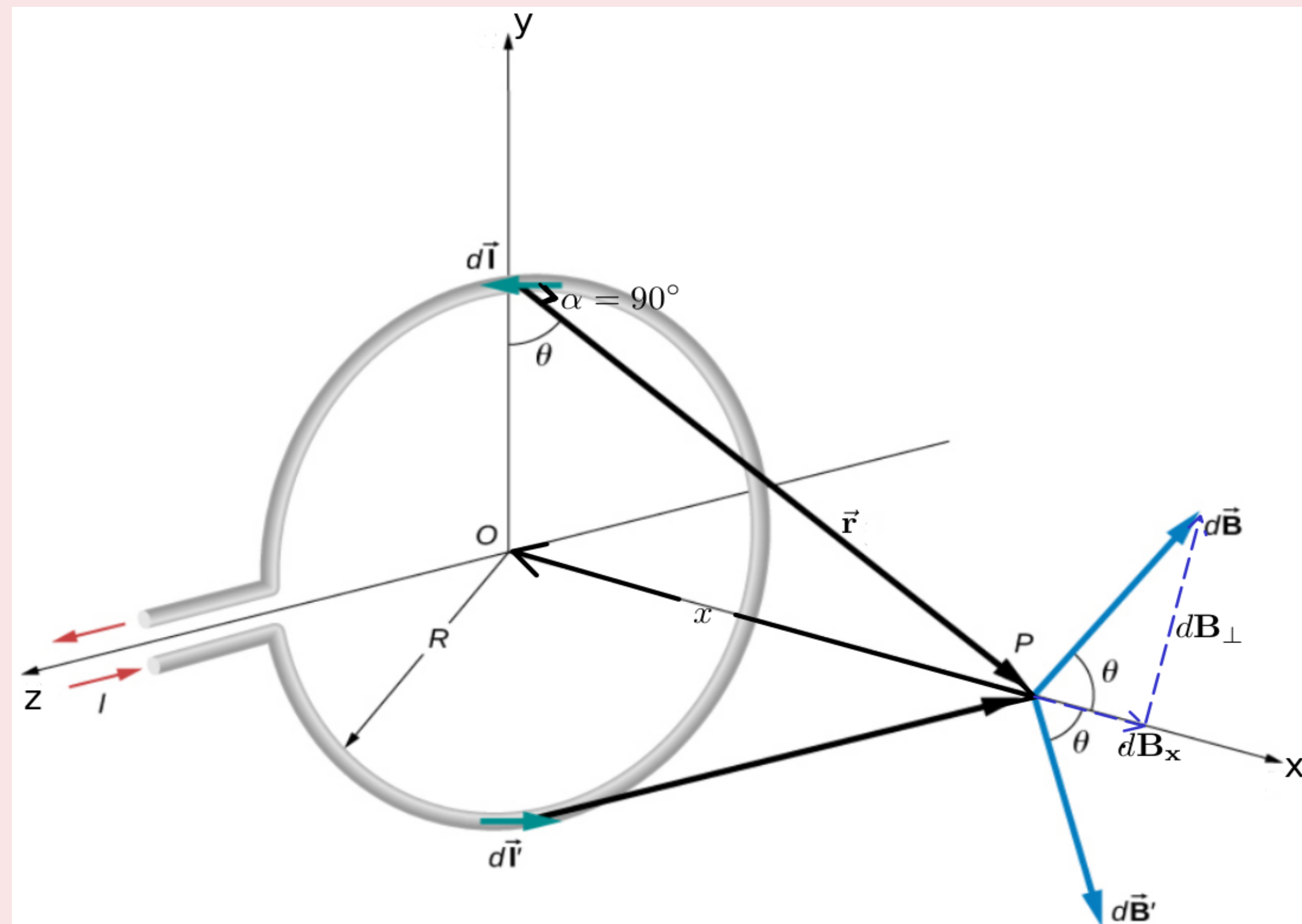


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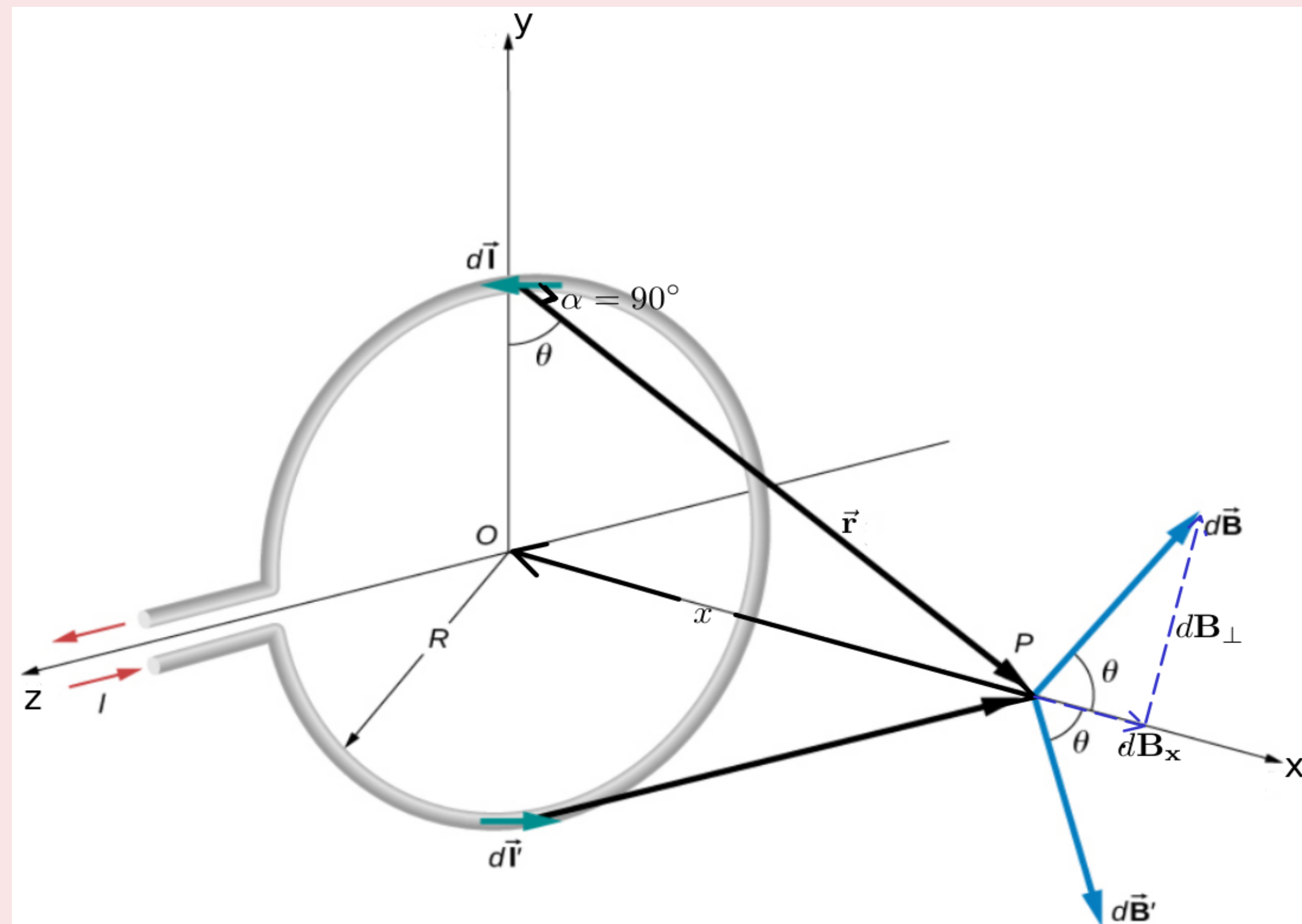


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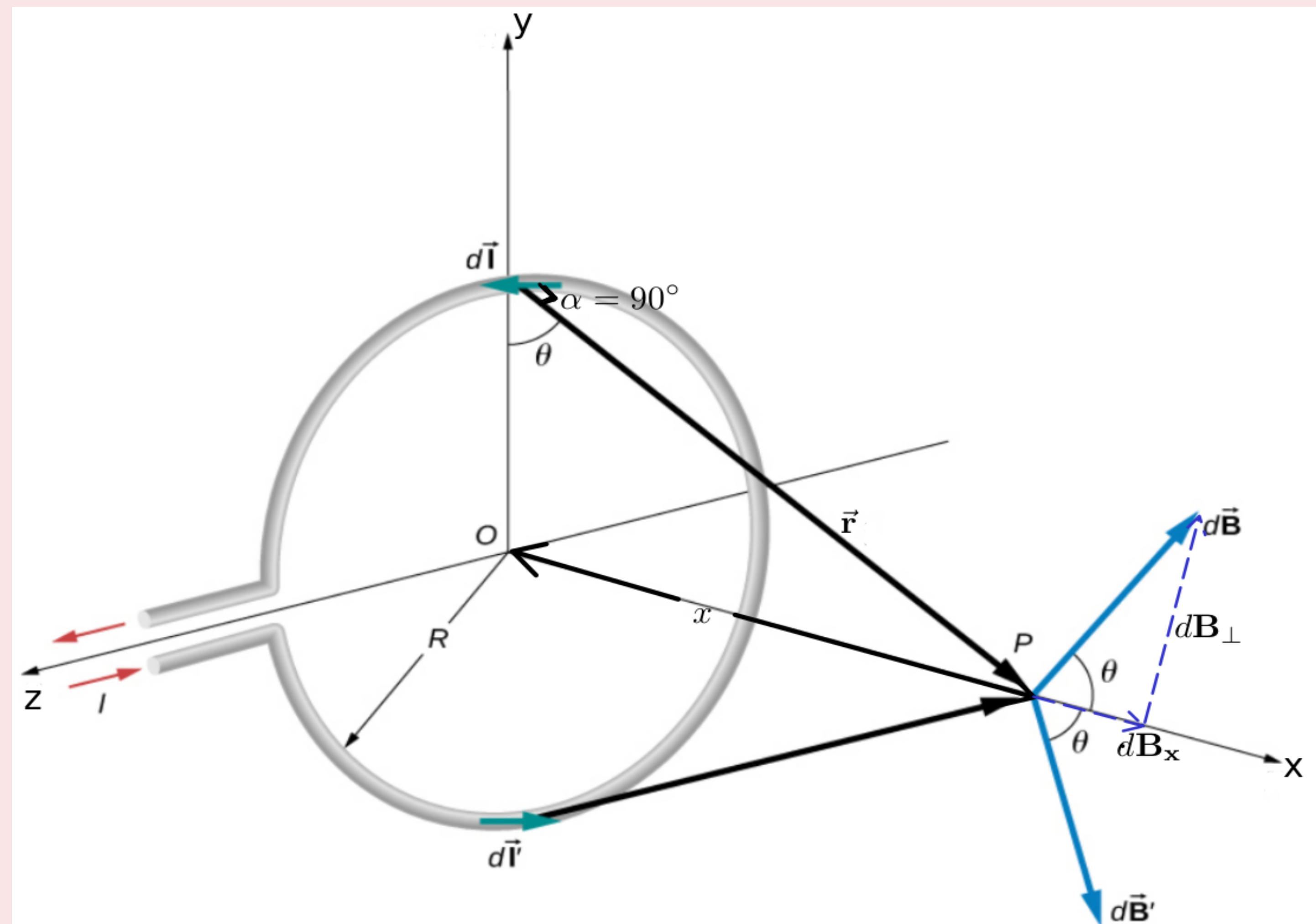


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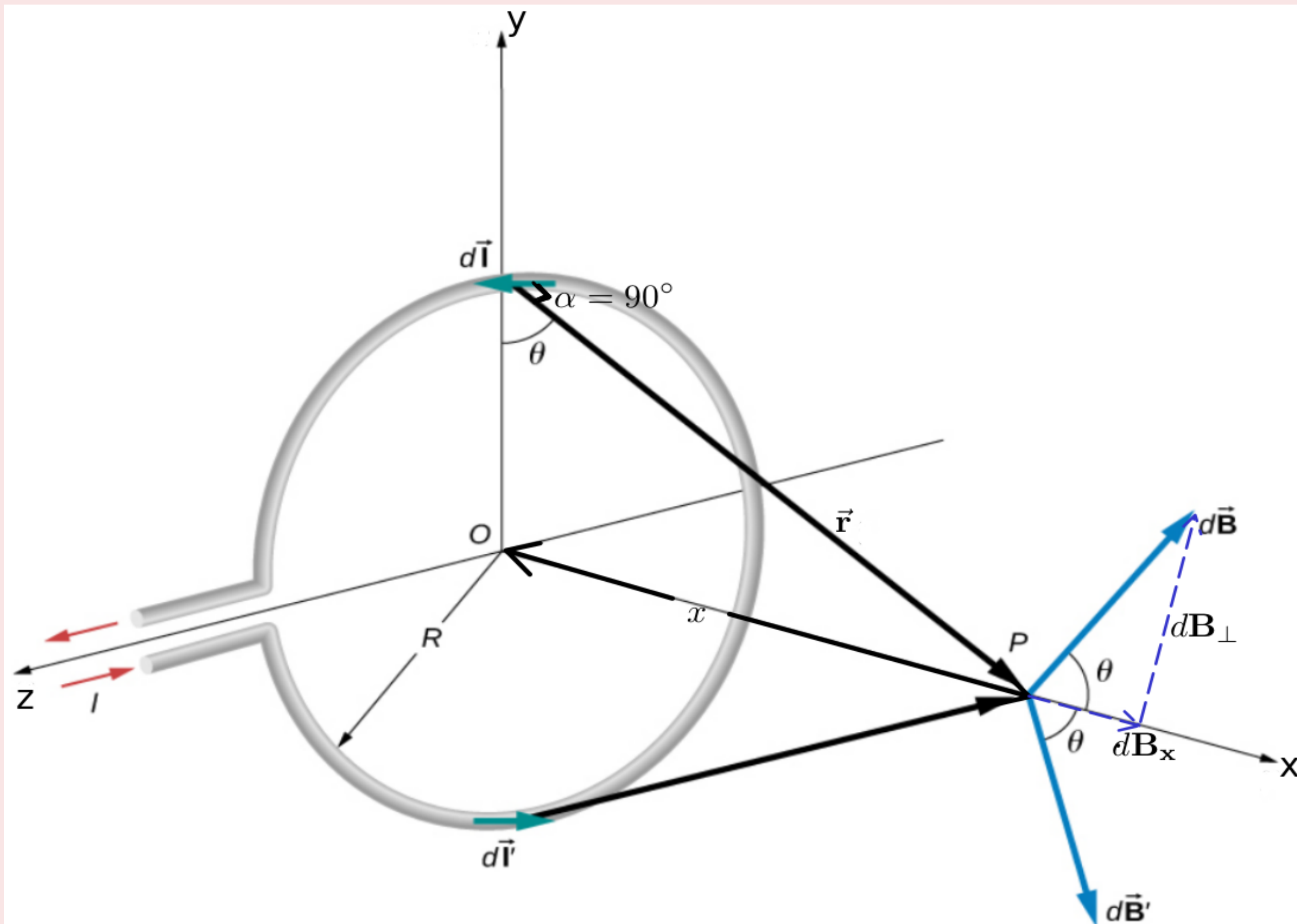


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- $d\vec{B}$  has an **x-component**  $dB_x$  and a **component perpendicular to x-axis**,  $d\vec{B}_\perp$ .



# Applications of Biot-Savart Law

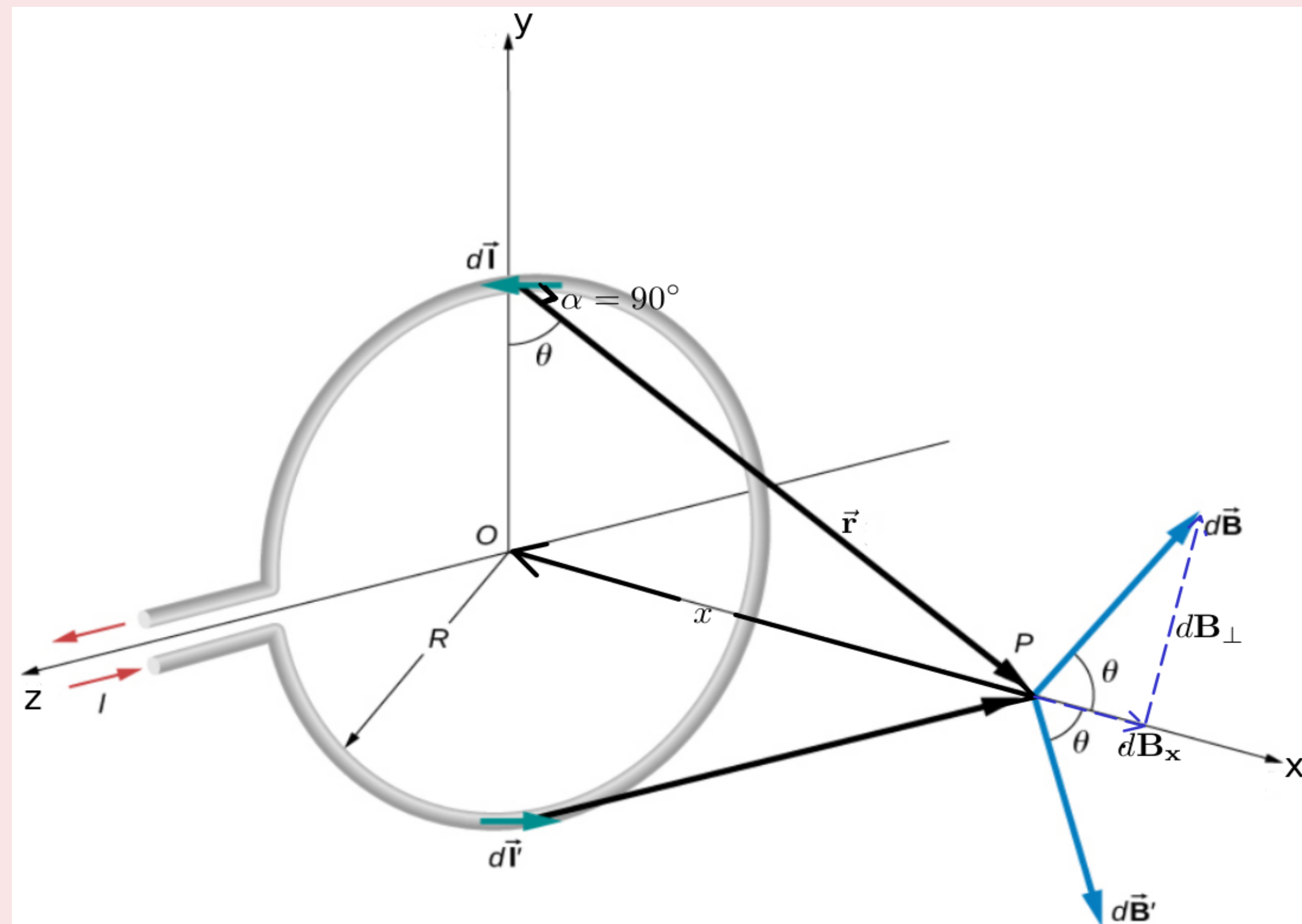


Figure 12.11 Determining the magnetic field at point P along the axis of a current-carrying loop of wire.

## Magnetic Field On The Axis Of a Circular Current Loop

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# Applications of Biot-Savart Law

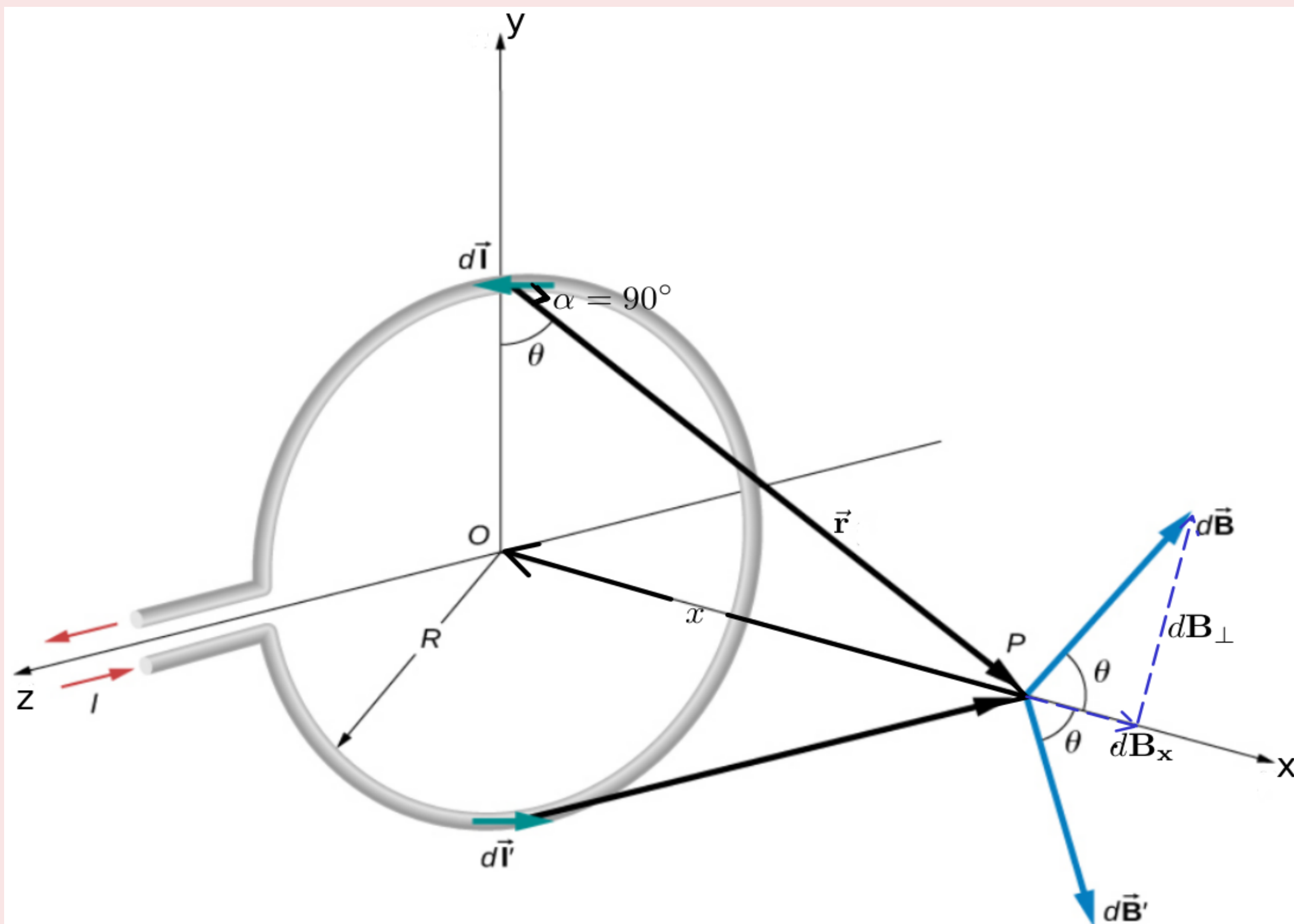


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- When the **perpendicular components** ( $dB_{\perp}$ ) are **summed over**, **they cancel out**.
- Thus, only the x-component ( $dB_x$ ) survives



# Applications of Biot-Savart Law

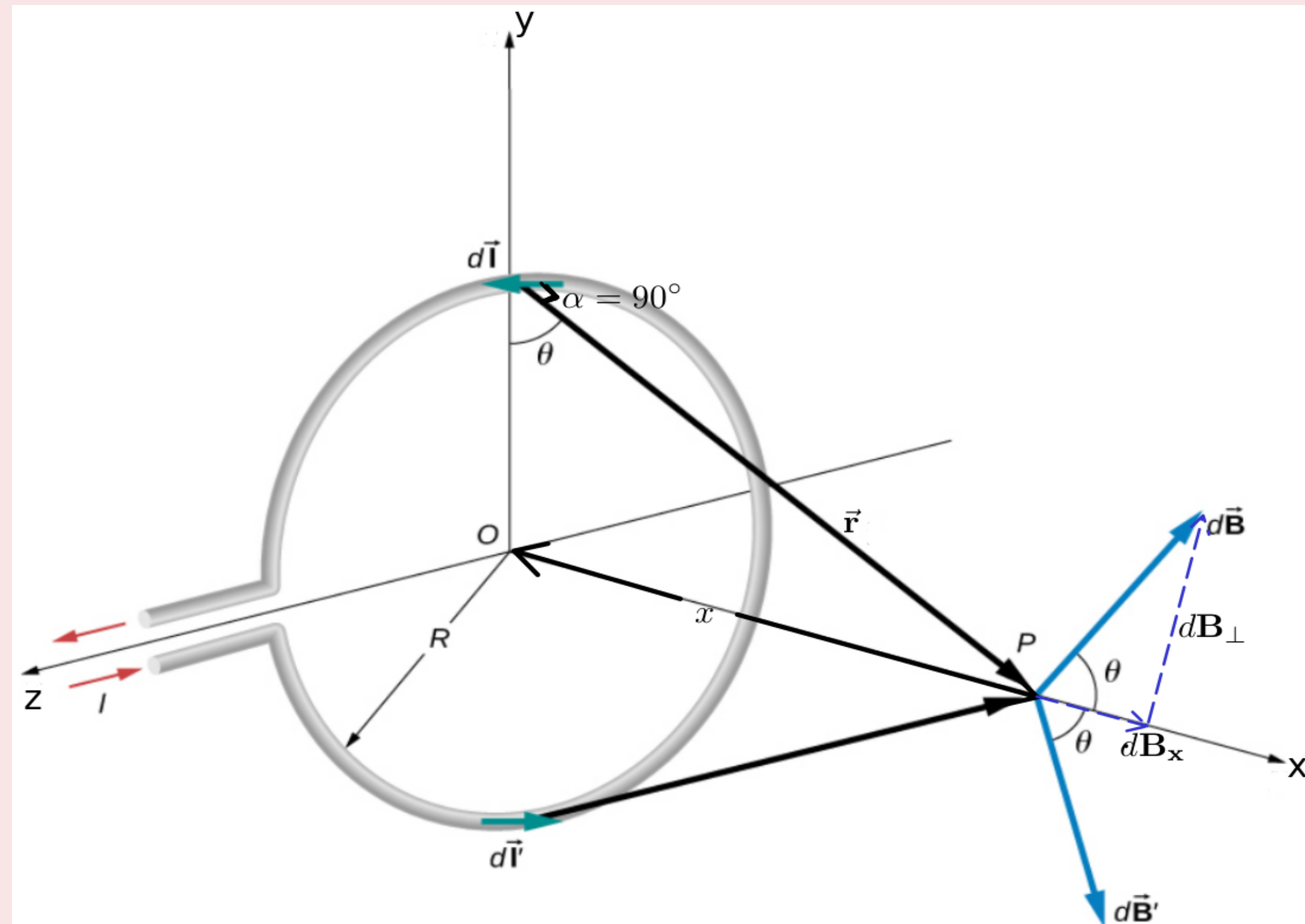


Figure 12.11 Determining the magnetic field at point  $P$  along the axis of a current-carrying loop of wire.

## Magnetic Field On The Axis Of a Circular Current Loop

- The **net contribution along x-direction** can be **obtained by integrating**  $dB_x = dB \cos \theta$  over the loop.



# Applications of Biot-Savart Law

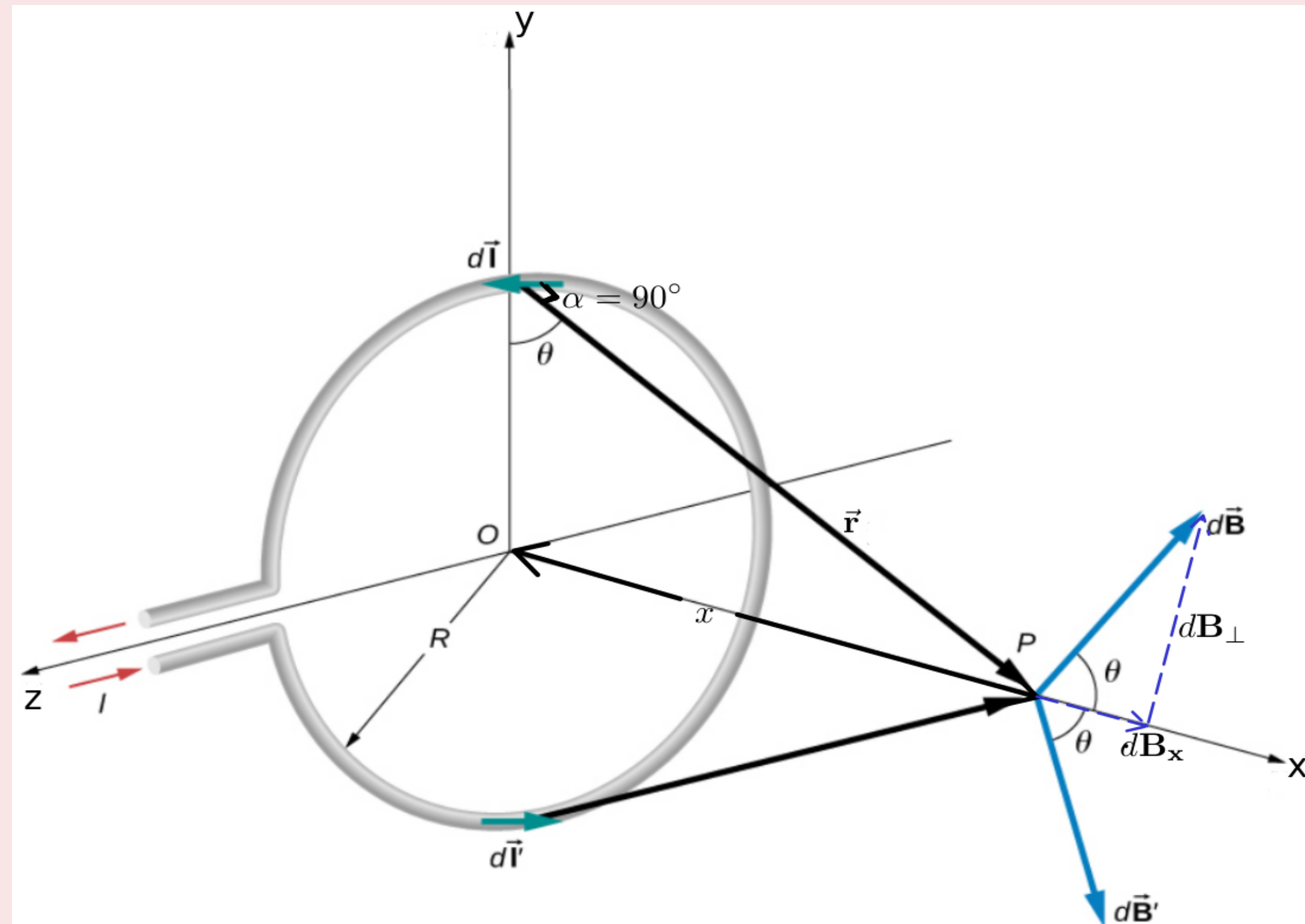


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$$\cos \theta = \frac{R}{(x^2 + R^2)^{\frac{1}{2}}} \quad \dots\dots(2)$$



# Applications of Biot-Savart Law

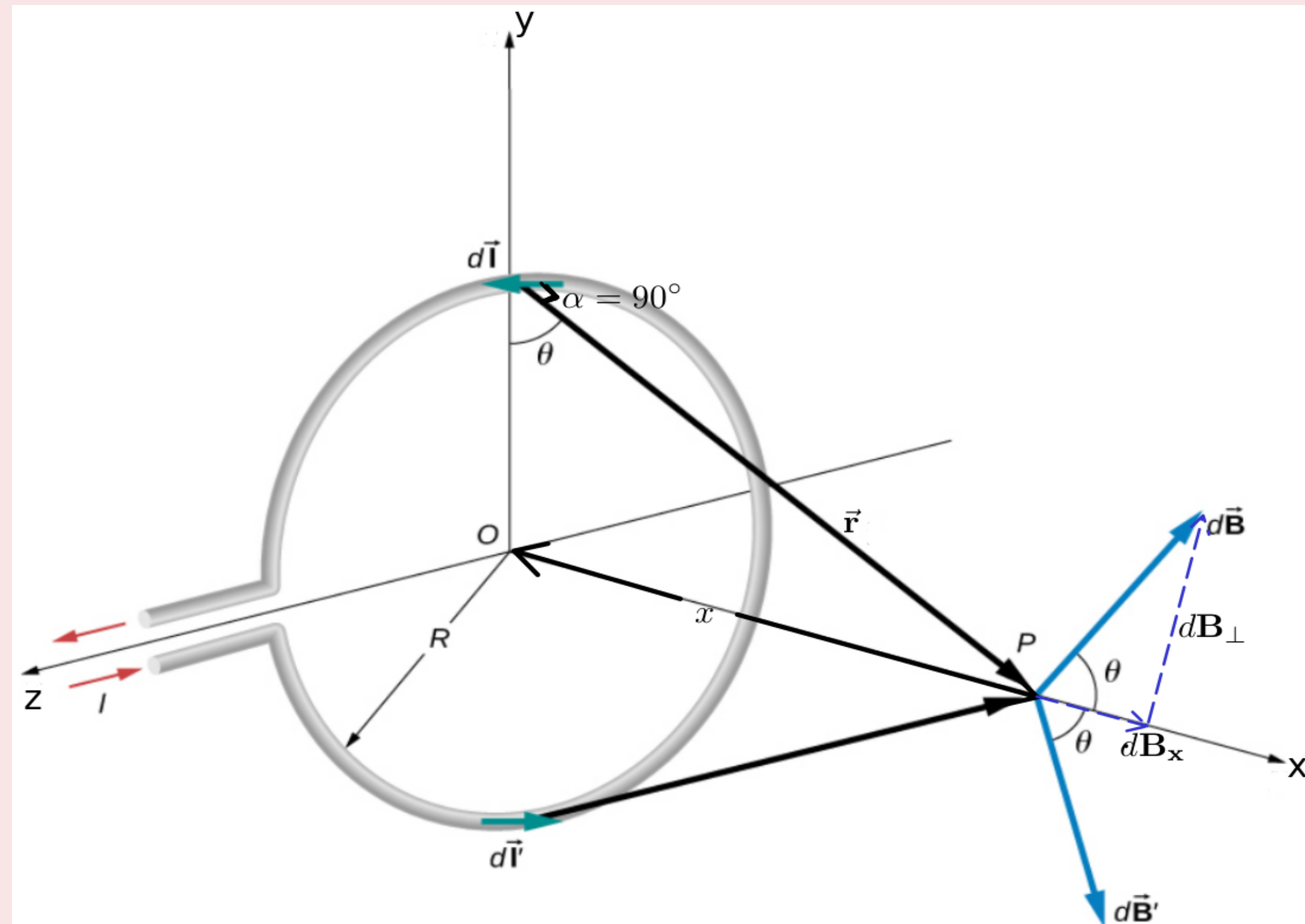


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- From Eqs. (1) and (2),

$$dB_x = \frac{\mu_0}{4\pi} \frac{Idl}{(x^2 + R^2)} \times \frac{R}{(x^2 + R^2)^{\frac{1}{2}}}$$



# Applications of Biot-Savart Law

## Magnetic Field On The Axis Of a Circular Current Loop

- $$dB_x = \frac{\mu_0}{4\pi} \frac{I R dl}{(x^2 + R^2)^{\frac{3}{2}}}$$



# Applications of Biot-Savart Law

## Magnetic Field On The Axis Of a Circular Current Loop

- $$dB_x = \frac{\mu_0}{4\pi} \frac{I R dl}{(x^2 + R^2)^{\frac{3}{2}}}$$

- Thus, the **magnetic field at P due to entire circular loop** is

$$\vec{B} = B_x \hat{i} = \int_0^{2\pi R} \frac{\mu_0}{4\pi} \frac{I R dl}{(x^2 + R^2)^{\frac{3}{2}}} \hat{i}$$



# Applications of Biot-Savart Law

## Magnetic Field On The Axis Of a Circular Current Loop

- $$dB_x = \frac{\mu_0}{4\pi} \frac{I R dl}{(x^2 + R^2)^{\frac{3}{2}}}$$

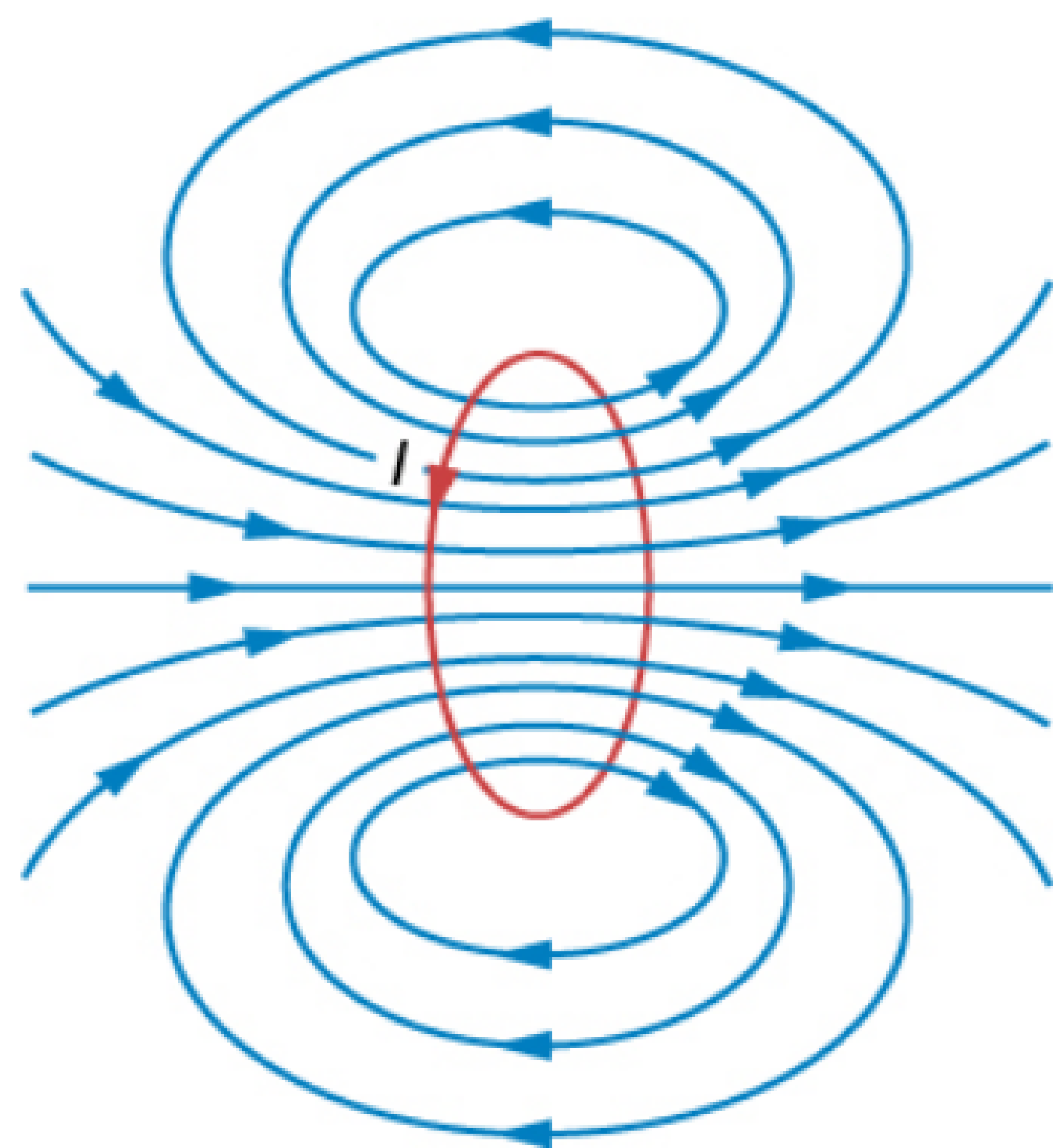
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$$\vec{B} = \frac{\mu_0}{4\pi} \frac{I R \times 2\pi R}{(x^2 + R^2)^{\frac{3}{2}}} \hat{i} \text{ Or } \boxed{\vec{B} = \frac{\mu_0}{2} \frac{I R^2}{(x^2 + R^2)^{\frac{3}{2}}} \hat{i}}$$



# Applications of Biot-Savart Law



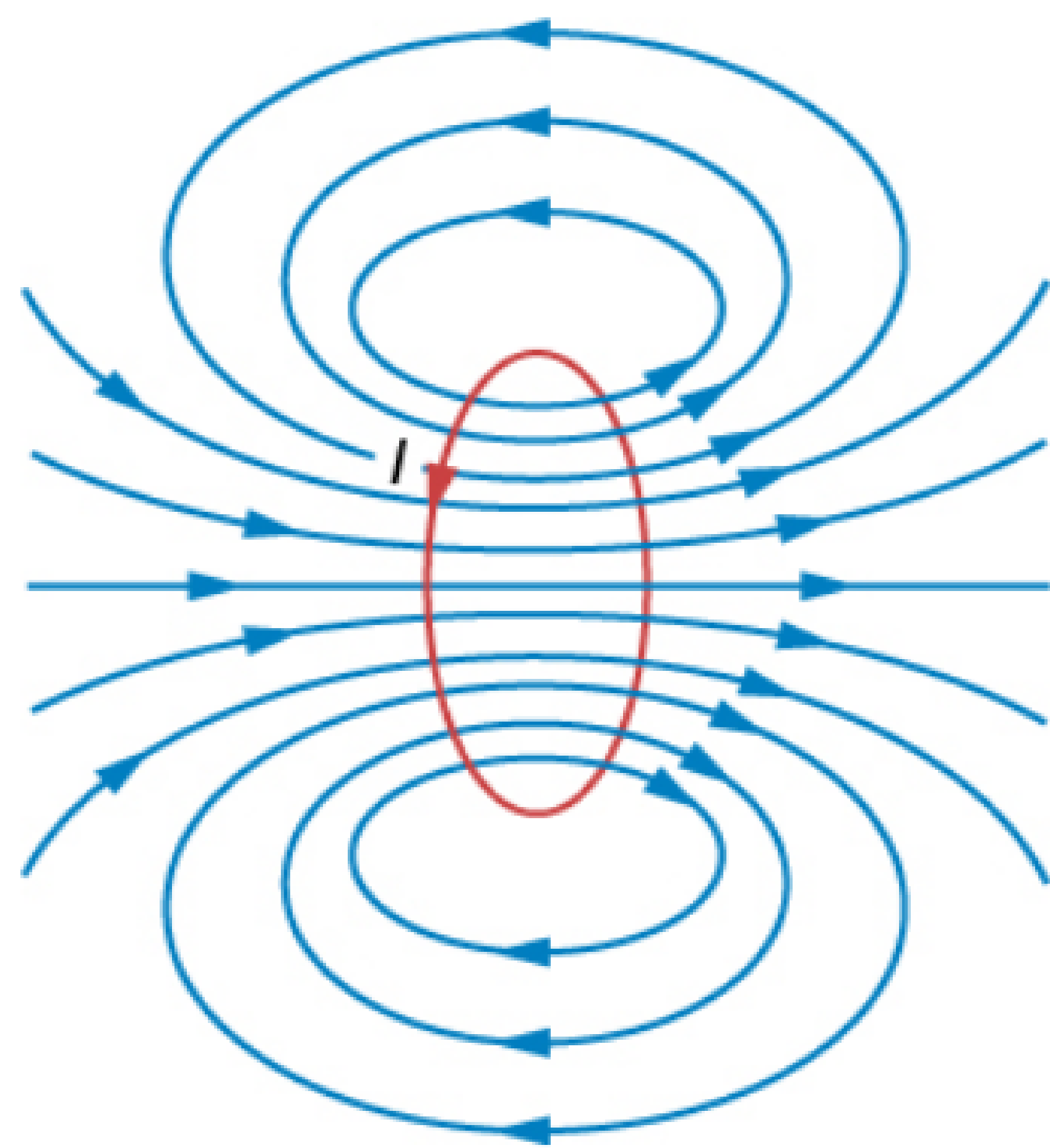
**Figure 12.12** Sketch of the magnetic field lines of a circular current loop.

## Magnetic Field On The Axis Of a Circular Current Loop

- **SPECIAL CASE:** The field at the center of the loop ( $x = 0$ ).



# Applications of Biot-Savart Law



**Figure 12.12** Sketch of the magnetic field lines of a circular current loop.

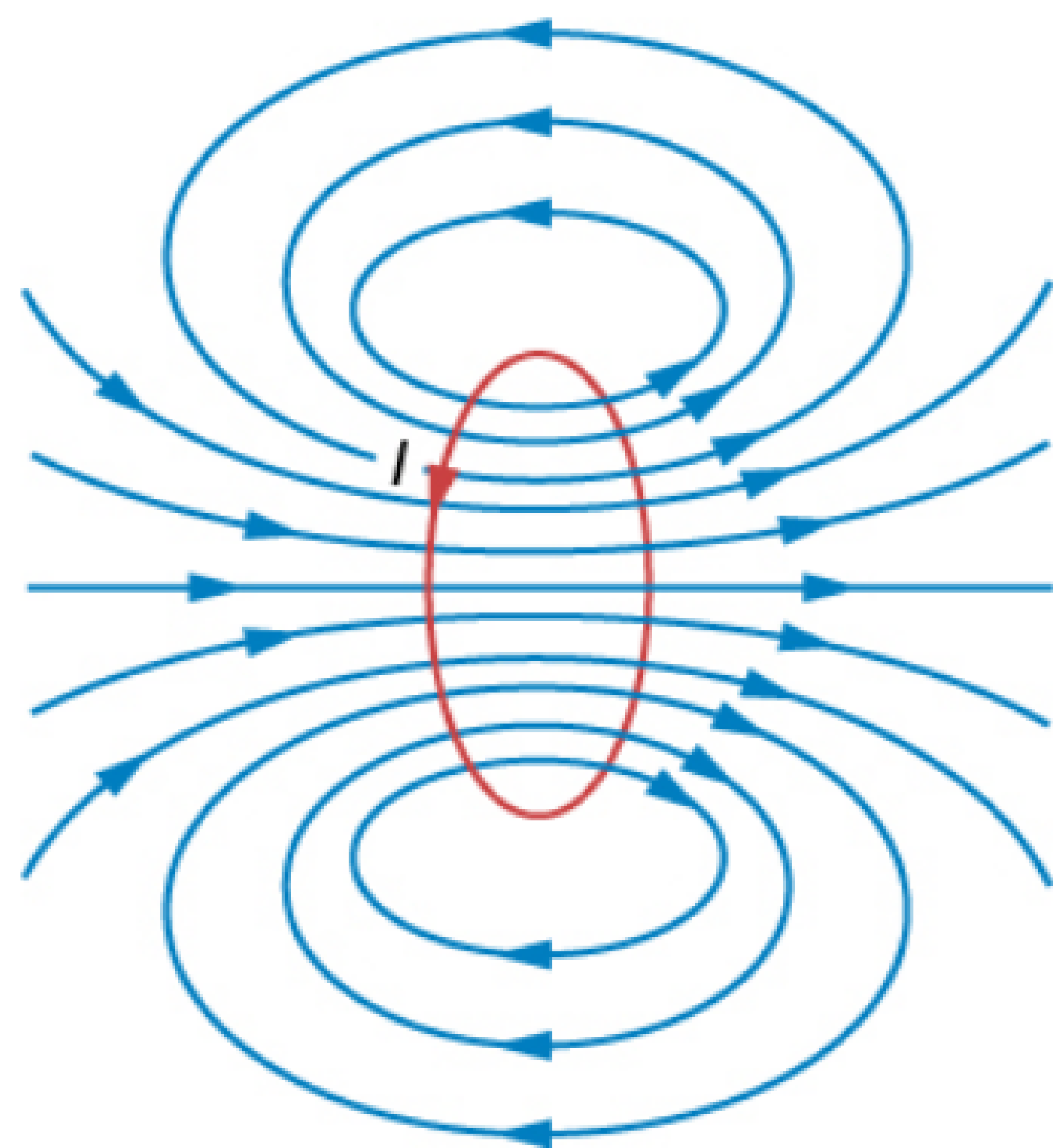
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$$\vec{B} = \frac{\mu_0}{2} \frac{I R^2}{(0 + R^2)^{\frac{3}{2}}} \hat{i}$$



# Applications of Biot-Savart Law



**Figure 12.12** Sketch of the magnetic field lines of a circular current loop.

## Magnetic Field On The Axis Of a Circular Current Loop

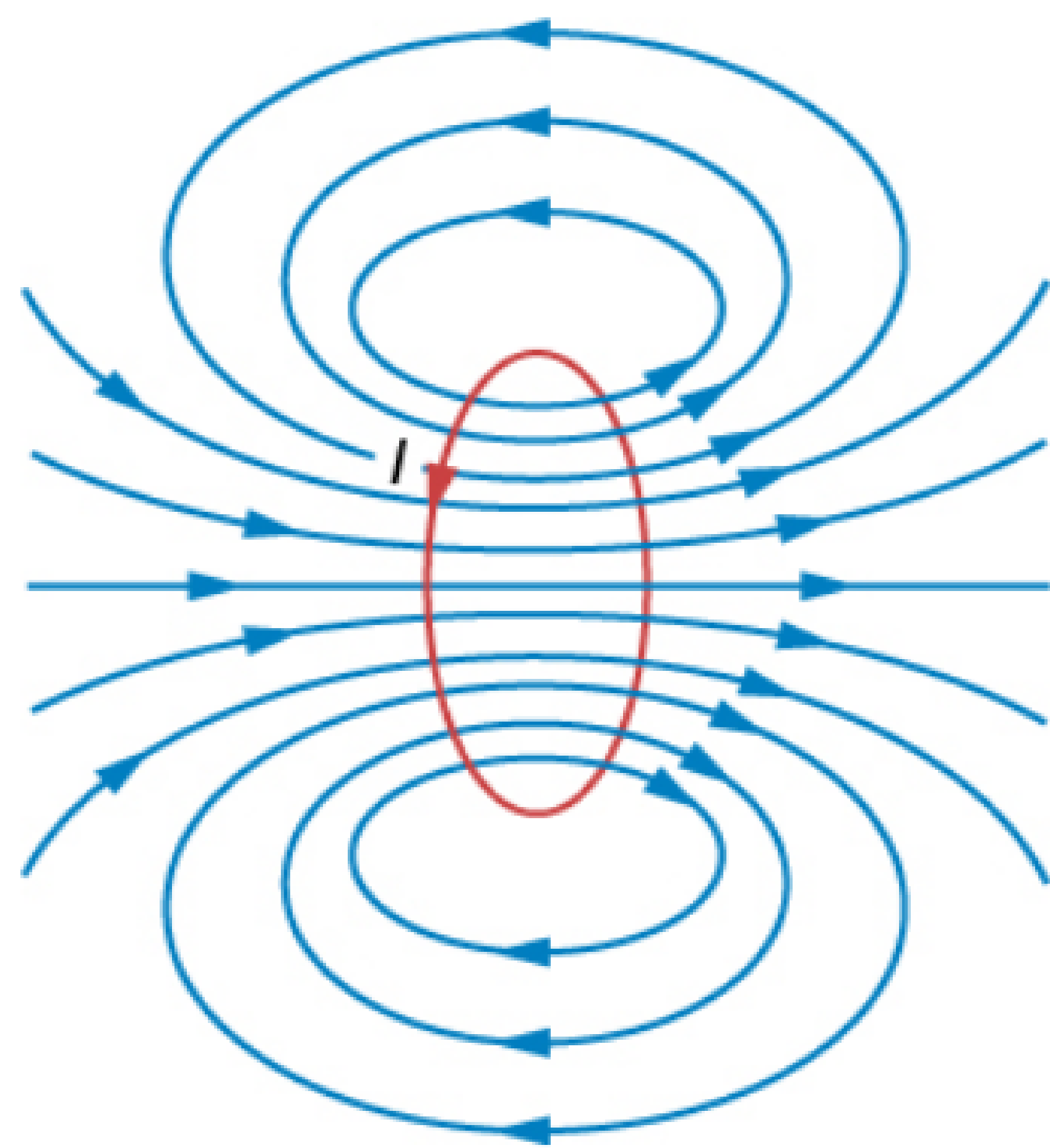
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# Applications of Biot-Savart Law



**Figure 12.12** Sketch of the magnetic field lines of a circular current loop.

## Magnetic Field On The Axis Of a Circular Current Loop

- **SPECIAL CASE:** The **field at the center of the loop** ( $x = 0$ ).

$$\vec{B} = \frac{\mu_0}{2} \frac{I R^2}{(0 + R^2)^{\frac{3}{2}}} \hat{i}$$

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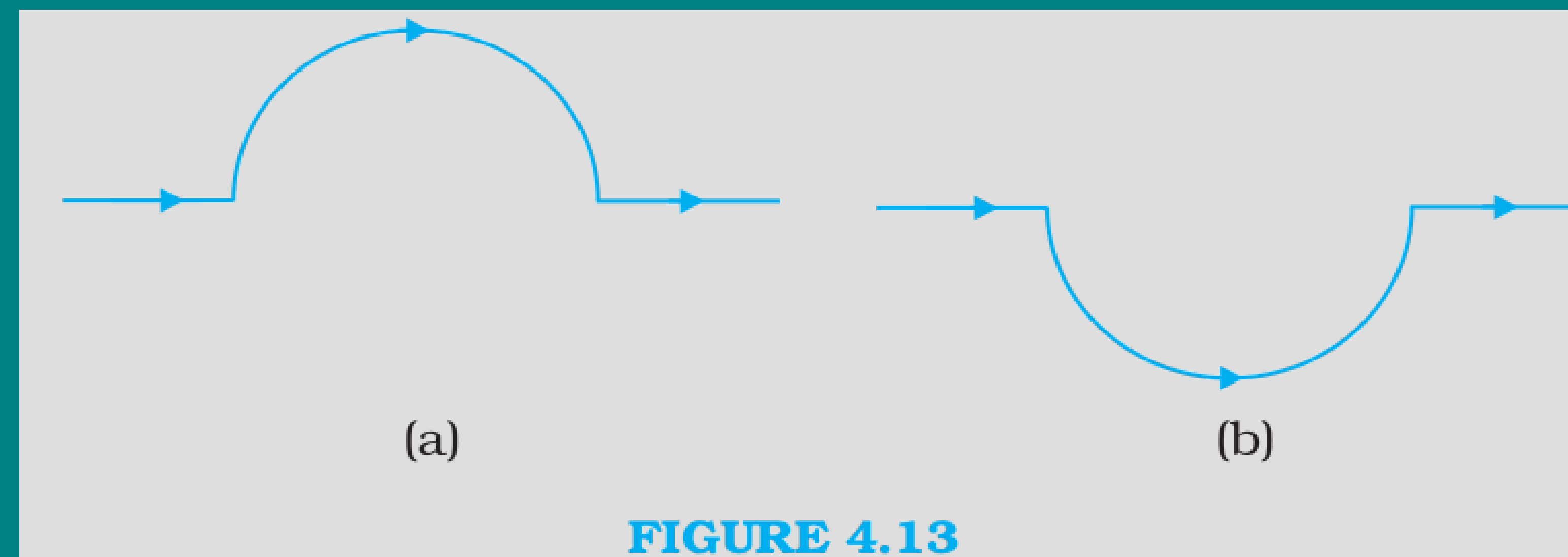
- The **magnetic field lines** due to a circular wire **form closed loops**



## Example 4.6

A straight wire carrying a **current of 12 A** is bent into a semi-circular arc of **radius 2.0 cm** as shown in Fig. 4.13(a). Consider the **magnetic field  $\mathbf{B}$**  at the **center** of the arc.

- (a) What is the **magnetic field due to the straight segments**?
- (b) In what way the **contribution to  $\mathbf{B}$**  from the **semicircle differs from that of a circular loop** and in what way does it resemble?
- (c) Would your answer be different **if the wire were bent** into a semi-circular arc of the same radius but **in the opposite way** as shown in Fig. 4.13(b)?





## Example 4.7

Consider a tightly wound **100 turn** coil of **radius 10 cm**, carrying a **current of 1 A**.  
**What is the magnitude of the magnetic field at the center of the coil?**